

modulo8

2026-07-28

```
paquetes <- c(
  "dplyr", "readr", "ggplot2", "tidyr",
  "tibble", "forcats", "scales", "janitor"
)

paquetes_faltan <- paquetes[!(paquetes %in% installed.packages()[,"Package"])]

if (length(paquetes_faltan) > 0) {
  install.packages(paquetes_faltan)
}

library(dplyr)
library(readr)
library(ggplot2)
library(tidyr)
library(tibble)
library(forcats)
library(scales)
library(janitor)

# 1. Importación de datos
# Se carga la base original del ESS6.
estudio <- read_csv("Datos/ESS6e02_7.csv", show_col_types = FALSE)

class(estudio)

## [1] "spec_tbl_df" "tbl_df"      "tbl"        "data.frame"

dim(estudio)

## [1] 54673  625

names(estudio)

## [1] "name"      "essround"  "edition"   "proddate"  "idno"      "cntry"
## [7] "dweight"   "pspwght"   "pweight"   "anweight"  "tvttot"    "tvpol"
## [13] "ppltrst"   "pplfair"   "pplhlp"    "polintr"   "trstlgl"    "trstplc"
## [19] "trstplt"   "trstep"    "trstun"    "trstprt"   "trstprl"    "vote"
## [25] "prvtal"    "prvtcbe"   "prvtcbg"   "prvtldch"  "prvtlacy"   "prvtlccz"
## [31] "prvtldel1" "prvtldel2" "prvtldcdk" "prvtlddee" "prvtldces"  "prvtldcfi"
## [37] "prvtldcfr" "prvtldgb"   "prvtldghu" "prvtldtaie" "prvtldtbl"  "prvtldtais"
## [43] "prvtldbit" "prvtldbt1" "prvtldbt2" "prvtldbt3" "prvtldtenl" "prvtldtano"
## [49] "prvtldcpl" "prvtldbpt" "prvtldcru" "prvtldtbse" "prvtldtsi"  "prvtldtcsk"
## [55] "prvtldcua" "prvtldtxk"  "prvtldtplt" "prvtldwrkprty" "prvtldwrkorg" "prvtldbadge"
## [61] "prvtldsgnptit" "prvtldpblm" "prvtldbctpr" "prvtldclsprty" "prvtldprtclal" "prvtldprtclcbe"
## [67] "prvtldprtclcbg" "prvtldprtclldch" "prvtldprtclldacy" "prvtldprtclldccz" "prvtldprtclldde" "prvtldprtclldcdk"
## [73] "prvtldprtcllddee" "prvtldprtclldces" "prvtldprtclldcfi" "prvtldprtclldcfr" "prvtldprtclldgb" "prvtldprtclldghu"
```

```

## [79] "prtlclcie" "prtlclcil" "prtlclais" "prtlclbit" "prtlclalt" "prtlclndl"
## [85] "prtlclano" "prtlclepl" "prtlclcpt" "prtlclcru" "prtlclbse" "prtlcldsi"
## [91] "prtlclcsk" "prtlcldua" "prtlclxk" "prtdgcl" "implvdm" "dmcntov"
## [97] "lrscale" "stflife" "stfecoc" "stfgov" "stfdem" "stfedu"
## [103] "stfhlt" "gincdif" "freehms" "euff" "imsmetn" "imdfetn"
## [109] "impcntr" "imbgeco" "imueclt" "imwbcnt" "happy" "sclmeet"
## [115] "inprdsc" "sclact" "crmvct" "aesfdrk" "health" "hlthhmp"
## [121] "rlgblg" "rlgdnm" "rlgdnal" "rlgdnbe" "rlgdnach" "rlgdncy"
## [127] "rlgdnade" "rlgdnafi" "rlgdngb" "rlgdnhu" "rlgdnie" "rlgdnil"
## [133] "rlgdnis" "rlgdnlit" "rlgdnnl" "rlgdnnno" "rlgdnapl" "rlgdnpt"
## [139] "rlgdnaru" "rlgdnase" "rlgdnsi" "rlgdnsk" "rlgdnu" "rlgblge"
## [145] "rlgdname" "rlgdeal" "rlgdebe" "rlgdeach" "rlgdecy" "rlgdeade"
## [151] "rlgdeafi" "rlgdegb" "rlgdehu" "rlgdeie" "rlgdeil" "rlgdeis"
## [157] "rlgdelt" "rlgdenl" "rlgdeno" "rlgdeapl" "rlgdept" "rlgdearu"
## [163] "rlgdease" "rlgdesi" "rlgdesk" "rlgdeua" "rlgdgr" "rlgatnd"
## [169] "pray" "dscrgrp" "dscrce" "dscrntn" "dscrllg" "dscrllng"
## [175] "dscretn" "dsccrge" "dsccrgnd" "dsccrsex" "dsccrdsb" "dsccroth"
## [181] "dsccrdk" "dsccrref" "dsccrnap" "dsccrna" "ctzcntr" "ctzshipc"
## [187] "brncntr" "cntbrthc" "livecnta" "lngthom1" "lngthom2" "blgetmg"
## [193] "facntr" "fbrncntb" "mocntr" "mbrncntb" "wkvlog" "optftr"
## [199] "pstvms" "flrms" "fltdpr" "flteeff" "slprl" "wrhpp"
## [205] "fltlnl" "enjl" "fltsd" "cldgng" "enrglot" "fltanx"
## [211] "fltpcfl" "dclvlf" "lchshcp" "accdng" "wrbkrm" "lrrntlf"
## [217] "pplahlp" "trtrsp" "dngval" "nhpftr" "lotsgot" "lfwrs"
## [223] "flclpla" "tmdotwa" "flapppl" "deaimpp" "tmimdng" "tmabndng"
## [229] "tmendng" "tnapsur" "sedirlf" "rehlppl" "prhlppl" "plinsoc"
## [235] "physact" "fairelc" "dspplvt" "dfprtalc" "oppcrgv" "medcrgv"
## [241] "meprinf" "rghmgpr" "votedir" "imvtctz" "cttres" "ctstogv"
## [247] "gtpelc" "gvctzpv" "gvexpdc" "grdfinc" "pltavie" "fairelcc"
## [253] "dspplvtc" "dfprtalc" "oppcrgvc" "medcrgvc" "meprinf" "rghmgprc"
## [259] "votedirc" "cttresac" "gtpelcc" "gvctzpv" "gvexpdccc" "grdfincc"
## [265] "pltaviec" "fplvdm" "fplvdmi" "fplvdmc" "pplvdmi" "pplvdmc"
## [271] "chpldm" "chpldmi" "chpldmc" "stpldmi" "stpldmc" "gvspcdm"
## [277] "gvspdmi" "gvspdmc" "gvcodmi" "gvcodmc" "hhmb" "gndr"
## [283] "gndr2" "gndr3" "gndr4" "gndr5" "gndr6" "gndr7"
## [289] "gndr8" "gndr9" "gndr10" "gndr11" "gndr12" "gndr13"
## [295] "gndr14" "gndr15" "gndr16" "gndr17" "gndr18" "gndr19"
## [301] "gndr20" "gndr21" "gndr22" "gndr23" "gndr24" "yrbrn"
## [307] "agea" "yrbrn2" "yrbrn3" "yrbrn4" "yrbrn5" "yrbrn6"
## [313] "yrbrn7" "yrbrn8" "yrbrn9" "yrbrn10" "yrbrn11" "yrbrn12"
## [319] "yrbrn13" "yrbrn14" "yrbrn15" "yrbrn16" "yrbrn17" "yrbrn18"
## [325] "yrbrn19" "yrbrn20" "yrbrn21" "yrbrn22" "yrbrn23" "yrbrn24"
## [331] "rshipa2" "rshipa3" "rshipa4" "rshipa5" "rshipa6" "rshipa7"
## [337] "rshipa8" "rshipa9" "rshipa10" "rshipa11" "rshipa12" "rshipa13"
## [343] "rshipa14" "rshipa15" "rshipa16" "rshipa17" "rshipa18" "rshipa19"
## [349] "rshipa20" "rshipa21" "rshipa22" "rshipa23" "rshipa24" "icpart1"
## [355] "rshpsts" "rshpsfi" "lvgtptnea" "dvrcdeva" "icpart2" "iccohbt"
## [361] "marsts" "marstfi" "marstgb" "marstie" "maritalb" "chldhm"
## [367] "chldhhe" "domicil" "eiscd" "edulvlb" "edlvdal" "edlvebe"
## [373] "edlvdbg" "edlvdc" "edlvdecy" "edlvdcz" "eduade1" "edude2"
## [379] "edude3" "edlvddk" "edlvdee" "edlvees" "edlvdfi" "edlvdf"
## [385] "eduagb1" "edugb2" "edagegb" "edlvdhu" "edlvdie" "eduail1"
## [391] "eduail2" "edlvdis" "edlvdit" "edlvdlit" "edlvdn1" "edlvdnno"
## [397] "edlvlepl" "edupl2" "edlvdpt" "edlvdr" "edlvdse" "edlvdsi"

```

```
## [403] "edlvnsk" "edlvdua" "edlvdxk" "eduyrs" "pdwrk" "edctn"
## [409] "uempla" "uempli" "dsbld" "rtrd" "cmsrv" "hswrk"
## [415] "dngoth" "dngref" "dngdk" "dngna" "icomdng" "mainact"
## [421] "mnactic" "icpdwrk" "crpdwk" "pdjobev" "pdjobyr" "emplrel"
## [427] "emplno" "wrkctra" "estsz" "jbspv" "njbspv" "wkdcorga"
## [433] "iorgact" "wkht" "wkhtot" "nacer2" "tporgwk" "isco08"
## [439] "wrkac6m" "icpdwk2" "stfjb" "stfjbot" "uemp3m" "uemp12m"
## [445] "uemp5yr" "mbtru" "hincsrca" "hinctnta" "hincfel" "icpart3"
## [451] "eiscdp" "edulvlpb" "edlvpdal" "edlvpebe" "edlvpdbg" "edlvpdch"
## [457] "edlvpecy" "edlvpcdz" "edupade1" "edupde2" "edupde3" "edlvpddk"
## [463] "edlvpeee" "edlvpees" "edlvpdfi" "edlvpdfr" "edupagb1" "edupgb2"
## [469] "edagepgb" "edlvpdhu" "edlvpdie" "edupail1" "edupail2" "edlvpdis"
## [475] "edlvpdit" "edlvpdlt" "edlvpdnl" "edlvpdno" "edlvpepl" "eduppl2"
## [481] "edlvpdpt" "edlvpdru" "edlvpdse" "edlvpdsi" "edlvpdsk" "edlvpdua"
## [487] "edlvpdxk" "pdwrkp" "edctnp" "uemplap" "uemplip" "dsbldp"
## [493] "rtrdp" "cmsrvp" "hswrkp" "dngothp" "dngnapp" "dngrefp"
## [499] "dngdkp" "dngnap" "icomdnp" "mnactp" "icppdwk" "crpdwkp"
## [505] "isco08p" "emprelp" "wkhtotp" "eiscdf" "edulvlfb" "edlvfdal"
## [511] "edlvfebe" "edlvfdbg" "edlvfdch" "edlvfecy" "edlvfdcz" "edufade1"
## [517] "edufde2" "edufde3" "edlvfddk" "edlvfdee" "edlvfees" "edlvfdfi"
## [523] "edlvfdfr" "edufagb1" "edufgb2" "edagefgb" "edlvfdhu" "edlvfdie"
## [529] "edufail1" "edufail2" "edlvfdis" "edlvfdit" "edlvfdlt" "edlvfdnl"
## [535] "edlvfdno" "edlvfepl" "edlvfdpt" "edlvfdru" "edlvfdse" "edlvfdsi"
## [541] "edlvfdsk" "edlvfdua" "edlvfdxk" "emprf14" "occf14b" "eiscedm"
## [547] "edulvlmb" "edlvmdal" "edlvmebe" "edlvmdbg" "edlvmdch" "edlvmeecy"
## [553] "edlvmdcz" "edumade1" "edumde2" "edumde3" "edlvmdck" "edlvmdde"
## [559] "edlvmees" "edlvmdfi" "edlvmdfr" "edumagb1" "edumgb2" "edagemgb"
## [565] "edlvmdhu" "edlvmdie" "edumail1" "edumail2" "edlvmdis" "edlvmdit"
## [571] "edlvmdlt" "edlvmdnl" "edlvmdno" "edlvmepl" "edlvmdpt" "edlvmdru"
## [577] "edlvmdse" "edlvmdsi" "edlvmdsk" "edlvmdua" "edlvmdxk" "emprf14"
## [583] "occm14b" "atncrse" "region" "regunit" "intewde" "ipcrtiv"
## [589] "imprich" "ipeqopt" "ipshabt" "impsafe" "impdiff" "ipfrule"
## [595] "ipudrst" "ipmodst" "ipgdtim" "impfree" "iphlppl" "ipsuces"
## [601] "ipstrgv" "ipadvnt" "ipbhprp" "iprspot" "iplylfr" "impenv"
## [607] "imptrad" "impfun" "inwdds" "inwmms" "inwyys" "inwshh"
## [613] "inwsmm" "inwdde" "inwmme" "inwyys" "inwehh" "inwemm"
## [619] "spltdme" "supqad1" "supqad2" "supqdd" "supqmm" "supqyr"
## [625] "inwtm"
```

2. Diccionario básico de variables clave

Este bloque documenta las variables seleccionadas para el análisis.

```
diccionario <- tribble(
  ~variable, ~pregunta, ~categorias,
  "gndr", "Gender", "1 = Male; 2 = Female; 9 = No answer",
  "icpart1", "Lives with husband/wife/partner", "1 = Yes; 2 = No; 9 = Not available",
  "chldhhe", "Ever had children living in household", "1 = Yes; 2 = No; 6 = Not applicable; 7 = Refusal",
  "dvrcdeva", "Ever been divorced/had civil union dissolved", "1 = Yes; 2 = No; 7 = Refusal; 8 = Don't know",
  "domicil", "Domicile, respondent's description", "1 = Big city; 2 = Suburbs/outskirts; 3 = Town/small town",
  "eiscdp", "Highest level of education", "1 to 7 = education levels; 55 = Other; 77 = Refusal; 88 = Don't know",
  "mnactic", "Main activity, last 7 days", "1 = Paid work; 2 = Education; 3 = Unemployed looking; 4 = Unemployed",
  "hincsrca", "Main source of household income", "1 = Wages; 2 = Self-employment; 3 = Farming; 4 = Pensions",
  "hincfel", "Feeling about household income nowadays", "1 = Living comfortably; 2 = Coping; 3 = Difficult",
  "cntry", "Country", "ISO country code"
```

```
)

diccionario

## # A tibble: 10 x 3
##   variable pregunta      categorias
##   <chr>      <chr>      <chr>
## 1 gndr      Gender      1 = Male; 2 = Female; ~
## 2 icpart1   Lives with husband/wife/partner 1 = Yes; 2 = No; 9 = N~
## 3 chldhhe   Ever had children living in household 1 = Yes; 2 = No; 6 = N~
## 4 dvrcdeva  Ever been divorced/had civil union dissolved 1 = Yes; 2 = No; 7 = R~
## 5 domicil   Domicile, respondent's description 1 = Big city; 2 = Subu~
## 6 eisced    Highest level of education 1 to 7 = education lev~
## 7 mnactic   Main activity, last 7 days 1 = Paid work; 2 = Edu~
## 8 hincsrca  Main source of household income 1 = Wages; 2 = Self-em~
## 9 hincfel   Feeling about household income nowadays 1 = Living comfortably~
## 10 cntry    Country      ISO country code
```

3. Exploración inicial de variables clave

Se revisan frecuencias y distribución de códigos antes de depurar.

```
vars_clave <- c("gndr", "icpart1", "chldhhe", "dvrcdeva", "domicil",
               "eisced", "mnactic", "hincsrca", "hincfel", "cntry")

for (v in vars_clave) {
  cat("\n===== \n")
  cat("Variable:", v, "\n")
  cat("Pregunta:", diccionario$pregunta[diccionario$variable == v], "\n")
  cat("Categorías:", diccionario$categorias[diccionario$variable == v], "\n\n")

  print(
    estudio %>%
      count(.data[[v]]) %>%
      mutate(porcentaje = round(100 * n / sum(n), 1))
  )
}
```

```
##
## =====
## Variable: gndr
## Pregunta: Gender
## Categorías: 1 = Male; 2 = Female; 9 = No answer
##
## # A tibble: 3 x 3
##   gndr      n porcentaje
##   <dbl> <int>      <dbl>
## 1     1 24929      45.6
## 2     2 29727      54.4
## 3     9   17         0
##
## =====
## Variable: icpart1
## Pregunta: Lives with husband/wife/partner
## Categorías: 1 = Yes; 2 = No; 9 = Not available
##
```

```

## # A tibble: 3 x 3
##   icpart1      n porcentaje
##   <dbl> <int>      <dbl>
## 1       1 31857      58.3
## 2       2 22603      41.3
## 3       9   213       0.4
##
## =====
## Variable: chldhhe
## Pregunta: Ever had children living in household
## Categorías: 1 = Yes; 2 = No; 6 = Not applicable; 7 = Refusal; 8 = Don't know; 9 = No answer
##
## # A tibble: 6 x 3
##   chldhhe      n porcentaje
##   <dbl> <int>      <dbl>
## 1       1 17504       32
## 2       2 15863       29
## 3       6 20945      38.3
## 4       7    76       0.1
## 5       8    36       0.1
## 6       9   249       0.5
##
## =====
## Variable: dvrcdeva
## Pregunta: Ever been divorced/had civil union dissolved
## Categorías: 1 = Yes; 2 = No; 7 = Refusal; 8 = Don't know; 9 = No answer
##
## # A tibble: 5 x 3
##   dvrcdeva      n porcentaje
##   <dbl> <int>      <dbl>
## 1       1  7891      14.4
## 2       2 46412      84.9
## 3       7   275       0.5
## 4       8    21        0
## 5       9    74       0.1
##
## =====
## Variable: domicil
## Pregunta: Domicile, respondent's description
## Categorías: 1 = Big city; 2 = Suburbs/outskirts; 3 = Town/small city; 4 = Country village; 5 = Farm/
##
## # A tibble: 8 x 3
##   domicil      n porcentaje
##   <dbl> <int>      <dbl>
## 1       1 12899      23.6
## 2       2  5682      10.4
## 3       3 16882      30.9
## 4       4 16256      29.7
## 5       5  2808       5.1
## 6       7     6        0
## 7       8    44       0.1
## 8       9    96       0.2
##
## =====

```

```

## Variable: eisced
## Pregunta: Highest level of education
## Categorías: 1 to 7 = education levels; 55 = Other; 77 = Refusal; 88 = Don't know; 99 = No answer
##
## # A tibble: 11 x 3
##   eisced      n porcentaje
##   <dbl> <int>      <dbl>
## 1      1  5897      10.8
## 2      2  9819       18
## 3      3  7353      13.4
## 4      4 12162      22.2
## 5      5  7348      13.4
## 6      6  5019       9.2
## 7      7  6711      12.3
## 8     55   186       0.3
## 9     77    30       0.1
## 10    88    75       0.1
## 11    99    73       0.1
##
## =====
## Variable: mnactic
## Pregunta: Main activity, last 7 days
## Categorías: 1 = Paid work; 2 = Education; 3 = Unemployed looking; 4 = Unemployed not looking; 5 = Si
##
## # A tibble: 12 x 3
##   mnactic      n porcentaje
##   <dbl> <int>      <dbl>
## 1      1 25760      47.1
## 2      2  4704       8.6
## 3      3  2978       5.4
## 4      4  1155       2.1
## 5      5  1291       2.4
## 6      6 12819      23.4
## 7      7    90       0.2
## 8      8  4796       8.8
## 9      9   520        1
## 10     77    39       0.1
## 11     88   105       0.2
## 12     99   416       0.8
##
## =====
## Variable: hincsrca
## Pregunta: Main source of household income
## Categorías: 1 = Wages; 2 = Self-employment; 3 = Farming; 4 = Pensions; 5 = Unemployment benefit; 6 =
##
## # A tibble: 11 x 3
##   hincsrca      n porcentaje
##   <dbl> <int>      <dbl>
## 1      1 30849      56.4
## 2      2  3007       5.5
## 3      3   813       1.5
## 4      4 14196       26
## 5      5  1359       2.5
## 6      6  1905       3.5

```

```
## 7      7  299      0.5
## 8      8  982      1.8
## 9     77  521      1
## 10     88  389     0.7
## 11     99  353     0.6
##
## =====
## Variable: hincfel
## Pregunta: Feeling about household income nowadays
## Categorías: 1 = Living comfortably; 2 = Coping; 3 = Difficult; 4 = Very difficult; 7 = Refusal; 8 =
##
## # A tibble: 7 x 3
##   hincfel      n porcentaje
##   <dbl> <int>      <dbl>
## 1      1 12785      23.4
## 2      2 23120      42.3
## 3      3 12157      22.2
## 4      4  5915      10.8
## 5      7   126       0.2
## 6      8   490       0.9
## 7      9    80       0.1
##
## =====
## Variable: cntry
## Pregunta: Country
## Categorías: ISO country code
##
## # A tibble: 29 x 3
##   cntry      n porcentaje
##   <chr> <int>      <dbl>
## 1 AL      1201       2.2
## 2 BE      1869       3.4
## 3 BG      2260       4.1
## 4 CH      1493       2.7
## 5 CY      1116        2
## 6 CZ      2009       3.7
## 7 DE      2958       5.4
## 8 DK      1650        3
## 9 EE      2380       4.4
## 10 ES      1889       3.5
## # i 19 more rows
```

4. Selección y depuración de variables

Se conservan solo las variables necesarias y se convierten a NA los códigos perdidos.

```
vars <- c(
  "idno", "cntry", "anweight", "happy", "sclmeet", "inprdsc", "sclact", "health",
  "pstvms", "fltdpr", "slprl", "enjl", "fltpcfl", "fltanx", "dclvlf", "accdng",
  "dngval", "tmdotwa", "sedirlf", "gndr", "yrbrn", "agea", "icpart1", "chldhhe",
  "dvrcdeva", "domicil", "eisced", "mnactic", "hincsrca", "hincfel"
)

df_dep <- estudio %>%
  select(any_of(vars)) %>%
```

```

mutate(
  across(c(happy),
    ~ replace(.x, .x %in% c(77, 88, 99), NA_real_)),
  across(c(sclmeet),
    ~ replace(.x, .x %in% c(77, 88, 99), NA_real_)),
  across(c(sclact, health, pstvms, fltdpr, slprl, enjlf,
    fltpcfl, fltanx, dclvlf, accdng, dngval, hincfel),
    ~ replace(.x, .x %in% c(7, 8, 9), NA_real_)),
  across(c(inprdsc, tmdotwa, sedirlf),
    ~ replace(.x, .x %in% c(77, 88, 99), NA_real_)),
  across(c(yrbrn),
    ~ replace(.x, .x %in% c(7777, 8888, 9999), NA_real_)),
  across(c(agea),
    ~ replace(.x, .x %in% c(999), NA_real_)),
  across(c(gndr, icpart1),
    ~ replace(.x, .x %in% c(9), NA_real_)),
  across(c(chldhhe),
    ~ replace(.x, .x %in% c(6, 7, 8, 9), NA_real_)),
  across(c(dvrcdeva),
    ~ replace(.x, .x %in% c(7, 8, 9), NA_real_)),
  across(c(domicil),
    ~ replace(.x, .x %in% c(7, 8, 9), NA_real_)),
  across(c(eisced),
    ~ replace(.x, .x %in% c(0, 55, 77, 88, 99), NA_real_)),
  across(c(mnactic),
    ~ replace(.x, .x %in% c(66, 77, 88, 99), NA_real_)),
  across(c(hincsrca),
    ~ replace(.x, .x %in% c(77, 88, 99), NA_real_))
)

write_csv(df_dep, "Datos/depurada/estudio_depurado.csv", na = "")

```

5. Recodificación para análisis e interpretación
Variables categóricas nominales/ordinales con etiquetas legibles.

```

df_rec <- df_dep %>%
  mutate(
    gndr_num = case_when(
      gndr == 1 ~ 1,
      gndr == 2 ~ 2,
      TRUE ~ NA_real_
    ),
    gndr = factor(
      gndr_num,
      levels = c(1, 2),
      labels = c("Hombre", "Mujer")
    ),

    icpart1_num = case_when(
      icpart1 == 1 ~ 1,
      icpart1 == 2 ~ 2,
      TRUE ~ NA_real_
    ),
    icpart1 = factor(

```



```

    icpart1_num,
    levels = c(1, 2),
    labels = c("Sí", "No")
  ),

  chldhhe_num = case_when(
    chldhhe == 1 ~ 1,
    chldhhe == 2 ~ 2,
    TRUE ~ NA_real_
  ),
  chldhhe = factor(
    chldhhe_num,
    levels = c(1, 2),
    labels = c("Sí", "No")
  ),

  dvrcdeva_num = case_when(
    dvrcdeva == 1 ~ 1,
    dvrcdeva == 2 ~ 2,
    TRUE ~ NA_real_
  ),
  dvrcdeva = factor(
    dvrcdeva_num,
    levels = c(1, 2),
    labels = c("Sí", "No")
  ),

  domicil = factor(
    domicil,
    levels = 1:5,
    labels = c(
      "Gran ciudad",
      "Suburbios o afueras",
      "Ciudad o pueblo pequeño",
      "Pueblo rural",
      "Granja o casa en el campo"
    )
  ),

  eisced = factor(
    eisced,
    levels = 1:7,
    labels = c(
      "Menos que secundaria inferior",      # 1
      "Secundaria inferior",                 # 2
      "Secundaria superior (nivel bajo)",    # 3
      "Secundaria superior (nivel alto)",    # 4
      "Formación avanzada / subgrado",      # 5
      "Terciaria inferior / grado",         # 6
      "Terciaria superior / máster+"        # 7
    )
  ),
  ordered = TRUE
),

```

```

mnactic = factor(
  mnactic,
  levels = 1:9,
  labels = c(
    "Trabajo remunerado",
    "Educación",
    "Desempleado busca trabajo",
    "Desempleado no busca trabajo",
    "Enfermo/discapacitado",
    "Jubilado",
    "Servicio comunitario/militar",
    "Tareas domésticas/cuidado",
    "Otro"
  )
),

hincsrca = factor(
  hincsrca,
  levels = 1:8,
  labels = c(
    "Salarios",
    "Autónomo",
    "Agricultura",
    "Pensiones",
    "Subsidio desempleo",
    "Otras ayudas sociales",
    "Ahorros/inversiones",
    "Otras fuentes"
  )
),

hincfel_num = case_when(
  hincfel == 1 ~ 4,
  hincfel == 2 ~ 3,
  hincfel == 3 ~ 2,
  hincfel == 4 ~ 1,
  TRUE ~ NA_real_
),

health_num = case_when(
  health == 1 ~ 5,
  health == 2 ~ 4,
  health == 3 ~ 3,
  health == 4 ~ 2,
  health == 5 ~ 1,
  TRUE ~ NA_real_
),

hincfel = factor(
  hincfel_num,
  levels = c(1, 2, 3, 4),
  labels = c("Mucha dificultad", "Dificultad", "Satisface", "Muy satisfecho"),
  ordered = TRUE
)

```

```

    ),

    health = factor(
      health_num,
      levels = c(1, 2, 3, 4, 5),
      labels = c("Muy mala", "Mala", "Regular", "Buena", "Muy buena"),
      ordered = TRUE
    )
  )
)

```

6. Comprobación rápida de la recodificación

```
table(df_rec$gndr, useNA = "ifany")
```

```
##
## Hombre  Mujer  <NA>
## 24929  29727    17

```

```
table(df_rec$gndr_num, useNA = "ifany")
```

```
##
##      1      2  <NA>
## 24929 29727    17

```

```
table(df_rec$icpart1, useNA = "ifany")
```

```
##
##      Sí      No  <NA>
## 31857 22603    213

```

```
table(df_rec$icpart1_num, useNA = "ifany")
```

```
##
##      1      2  <NA>
## 31857 22603    213

```

```
table(df_rec$chldhhe, useNA = "ifany")
```

```
##
##      Sí      No  <NA>
## 17504 15863 21306

```

```
table(df_rec$chldhhe_num, useNA = "ifany")
```

```
##
##      1      2  <NA>
## 17504 15863 21306

```

```
table(df_rec$dvrdeva, useNA = "ifany")
```

```
##
##      Sí      No  <NA>
##  7891 46412   370

```

```
table(df_rec$dvrdeva_num, useNA = "ifany")
```

```
##
##      1      2  <NA>
##  7891 46412   370

```

7. Resumen de variables numéricas y ordinales

```
vars_num <- c(
  "agea", "happy", "sclmeet", "inprdsc", "sclact",
  "pstvms", "fltdpr", "slprl", "enjlif", "fltpcfl", "fltanx",
  "dclvlf", "accdng", "dngval", "tmdotwa", "sedirlf",
  "health_num", "hincfel_num"
)

resumen_num <- lapply(vars_num, function(v) {
  x <- as.numeric(df_rec[[v]])
  data.frame(
    variable = v,
    n = sum(!is.na(x)),
    minimo = min(x, na.rm = TRUE),
    maximo = max(x, na.rm = TRUE),
    mediana = median(x, na.rm = TRUE),
    media = round(mean(x, na.rm = TRUE), 2),
    sd = round(sd(x, na.rm = TRUE), 2)
  )
}) %>% bind_rows()
```

resumen_num

##	variable	n	minimo	maximo	mediana	media	sd
## 1	agea	54540	15	103	48	48.31	18.59
## 2	happy	54228	0	10	8	7.11	2.10
## 3	sclmeet	54351	1	7	5	4.79	1.63
## 4	inprdsc	53753	0	6	3	2.67	1.44
## 5	sclact	53116	1	5	3	2.67	0.97
## 6	pstvms	54275	1	5	2	2.12	0.81
## 7	fltdpr	54294	1	4	1	1.52	0.71
## 8	slprl	54389	1	4	2	1.79	0.85
## 9	enjlif	54011	1	4	3	2.86	0.86
## 10	fltpcfl	54132	1	4	3	2.79	0.83
## 11	fltanx	54131	1	4	2	1.66	0.76
## 12	dclvlf	54395	1	5	2	1.99	0.90
## 13	accdng	54122	1	5	2	2.24	0.86
## 14	dngval	54058	1	5	2	2.06	0.78
## 15	tmdotwa	54062	0	10	7	6.63	2.23
## 16	sedirlf	53598	0	10	7	7.00	2.20
## 17	health_num	54573	1	5	4	3.75	0.94
## 18	hincfel_num	53977	1	4	3	2.79	0.93

```
write.csv(resumen_num, "Datos/depurada/tarea1_resumen_numericas.csv", row.names = FALSE)
```

8. Resumen de variables categóricas

```
vars_cat <- c("gnldr", "icpart1", "chldhhe", "dvrcdeva", "domicil",
  "eisced", "mnactic", "hincsrca", "cntry")

resumen_cat <- lapply(vars_cat, function(v) {
  x <- df_rec[[v]]
  data.frame(
```

```

    variable = v,
    n = sum(!is.na(x)),
    niveles = dplyr::n_distinct(x, na.rm = TRUE)
  )
}) %>% bind_rows()

resumen_cat

##   variable      n niveles
## 1      gndr 54656         2
## 2    icpart1 54460         2
## 3   chldhhe 33367         2
## 4  dvrccdeva 54303         2
## 5   domicil 54527         5
## 6    eisced 54309         7
## 7   mnactic 54113         9
## 8  hincsrca 53410         8
## 9     cntry 54673        29

write.csv(resumen_cat, "Datos/depurada/tarea1_resumen_categoricas.csv", row.names = FALSE)

# TAREA 1. EDA bottom-up
# Descripción, distribuciones, outliers, relaciones y subgrupos
# =====

# -----
# 9.0 Paquetes
# -----

paquetes <- c("dplyr", "tidyr", "ggplot2", "forcats", "scales", "readr", "purrr", "stringr")

instalar <- paquetes[!paquetes %in% installed.packages()[, "Package"]]
if (length(instalar) > 0) install.packages(instalar)

# Paquetes necesarios
paquetes <- c(
  "dplyr", "tidyr", "ggplot2", "forcats",
  "scales", "readr", "purrr", "stringr"
)

# Instalar los que faltan
paquetes_faltan <- paquetes[!(paquetes %in% installed.packages()[, "Package"])]
if (length(paquetes_faltan) > 0) {
  install.packages(paquetes_faltan)
}

# Cargar todos
invisible(lapply(paquetes, library, character.only = TRUE))

##
## Attaching package: 'purrr'

## The following object is masked from 'package:scales':
##
##   discard

```

```

# -----
# 9.1 Tema gráfico global
# Inspiración: Data-to-Viz / R Graph Gallery
# -----

theme_set(
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold", size = 14, colour = "#1F2A44"),
    plot.subtitle = element_text(size = 11, colour = "#5B657A"),
    strip.text = element_text(face = "bold", colour = "#1F2A44"),
    axis.title = element_text(face = "bold", colour = "#1F2A44"),
    axis.text = element_text(colour = "#3E4A59"),
    panel.grid.minor = element_blank(),
    panel.grid.major.x = element_blank(),
    legend.position = "right",
    legend.title = element_text(face = "bold"),
    plot.title.position = "plot"
  )
)

dir.create("Datos/depurada", recursive = TRUE, showWarnings = FALSE)
dir.create("Graficos", recursive = TRUE, showWarnings = FALSE)

```

```

# -----
# 9.2 Definición de variables y etiquetas
# -----

vars_num <- c(
  "agea", "happy", "sclmeet", "inprdsc", "sclact",
  "pstvms", "fltdpr", "slprl", "enjl", "fltpcfl", "fltanx",
  "dclvlf", "accdng", "dngval", "tmdotwa", "sedirlf",
  "health_num", "hincfel_num"
)

vars_cat <- c(
  "gnr", "icpart1", "chldhhe", "dvrcdeva",
  "domicil", "eiscd", "mnactic", "hincsrca", "cntry"
)

vars_vd_dashboard_num <- c(
  "happy", "sclmeet", "inprdsc", "dclvlf",
  "tmdotwa", "sedirlf", "health_num", "hincfel_num"
)

etiquetas_num <- c(
  agea = "Edad",
  happy = "Felicidad",
  sclmeet = "Frecuencia de reuniones sociales",
  inprdsc = "Personas íntimas",
  sclact = "Actividad social",
  pstvms = "Estado de ánimo positivo",
  fltdpr = "Depresión",
  slprl = "Problemas de sueño",

```

```

enjlfr = "Disfrute de la vida",
fltpcfl = "Sentirse capaz",
fltanx = "Ansiedad",
dclvlf = "Libertad para decidir",
accdng = "Logro",
dngval = "Valoración",
tmdotwa = "Tiempo para uno mismo",
sdirflr = "Dirección vital",
health_num = "Salud percibida",
hincfel_num = "Percepción ingresos hogar"
)

etiquetas_vi <- c(
  gndr = "Sexo",
  icpart1 = "Convivencia en pareja",
  chldhhe = "Niños en el hogar",
  dvrdeva = "Antecedente de divorcio",
  eisced = "Nivel educativo",
  hincfel = "Percepción de ingresos del hogar"
)

df_viz <- df_rec %>%
  mutate(
    eisced = factor(
      eisced,
      levels = levels(df_rec$eisced),
      labels = levels(df_rec$eisced),
      ordered = TRUE
    )
  )

```

9.3 Resumen descriptivo de variables

```

# -----

resumen_num <- lapply(vars_num, function(v) {
  x <- as.numeric(df_viz[[v]])
  data.frame(
    variable = v,
    etiqueta = etiquetas_num[[v]],
    n = sum(!is.na(x)),
    minimo = min(x, na.rm = TRUE),
    maximo = max(x, na.rm = TRUE),
    mediana = median(x, na.rm = TRUE),
    media = round(mean(x, na.rm = TRUE), 2),
    sd = round(sd(x, na.rm = TRUE), 2)
  )
}) %>% bind_rows()

resumen_cat <- lapply(vars_cat, function(v) {
  x <- df_viz[[v]]
  data.frame(
    variable = v,
    n = sum(!is.na(x)),
    niveles = dplyr::n_distinct(x, na.rm = TRUE)
  )
}) %>% bind_rows()

```

```

    )
  }) %>% bind_rows()

write.csv(resumen_num, "Datos/depurada/tareal_resumen_numericas.csv", row.names = FALSE)
write.csv(resumen_cat, "Datos/depurada/tareal_resumen_categoricas.csv", row.names = FALSE)

# 9.4 Distribuciones univariadas y outliers
# -----

df_num_long <- df_viz %>%
  select(all_of(c("agea", vars_vd_dashboard_num))) %>%
  pivot_longer(
    cols = everything(),
    names_to = "variable",
    values_to = "valor"
  ) %>%
  filter(!is.na(valor)) %>%
  mutate(
    variable = recode(variable, !!!etiquetas_num),
    variable = factor(variable, levels = unname(etiquetas_num[c("agea", vars_vd_dashboard_num)]))
  )

g_dist <- ggplot(df_num_long, aes(x = valor)) +
  geom_histogram(
    bins = 20,
    fill = "#4C78A8",
    color = "white",
    alpha = 0.9
  ) +
  facet_wrap(~ variable, scales = "free", ncol = 3) +
  labs(
    title = "Distribución de variables numéricas y ordinales resumidas",
    subtitle = "Exploración de forma, concentración y asimetría",
    x = "Valor",
    y = "Frecuencia"
  )

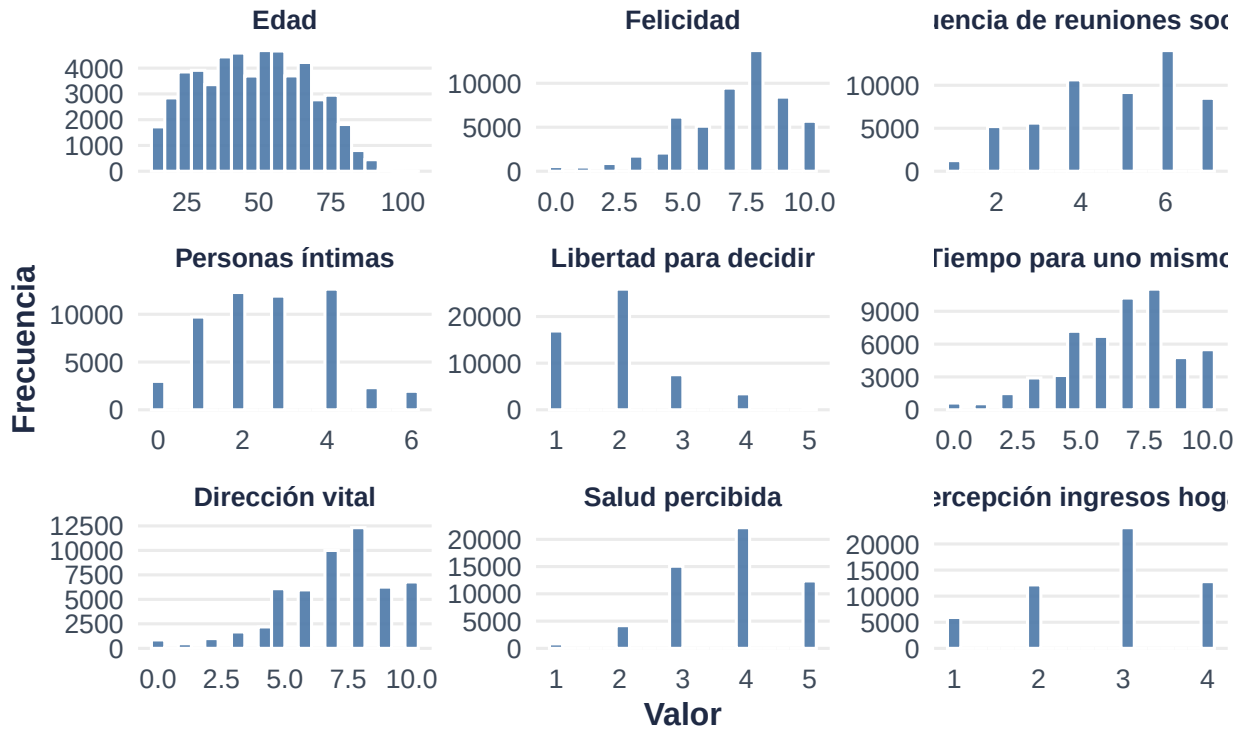
g_outliers <- ggplot(df_num_long, aes(x = variable, y = valor, fill = variable)) +
  geom_boxplot(alpha = 0.85, outlier.color = "#E45756", outlier.alpha = 0.25) +
  coord_flip() +
  scale_fill_brewer(palette = "Set3") +
  labs(
    title = "Valores atípicos y dispersión",
    subtitle = "Boxplots de variables clave",
    x = NULL,
    y = "Valor"
  ) +
  theme(legend.position = "none")

g_dist

```


Distribución de variables numéricas y ordinales resumidas

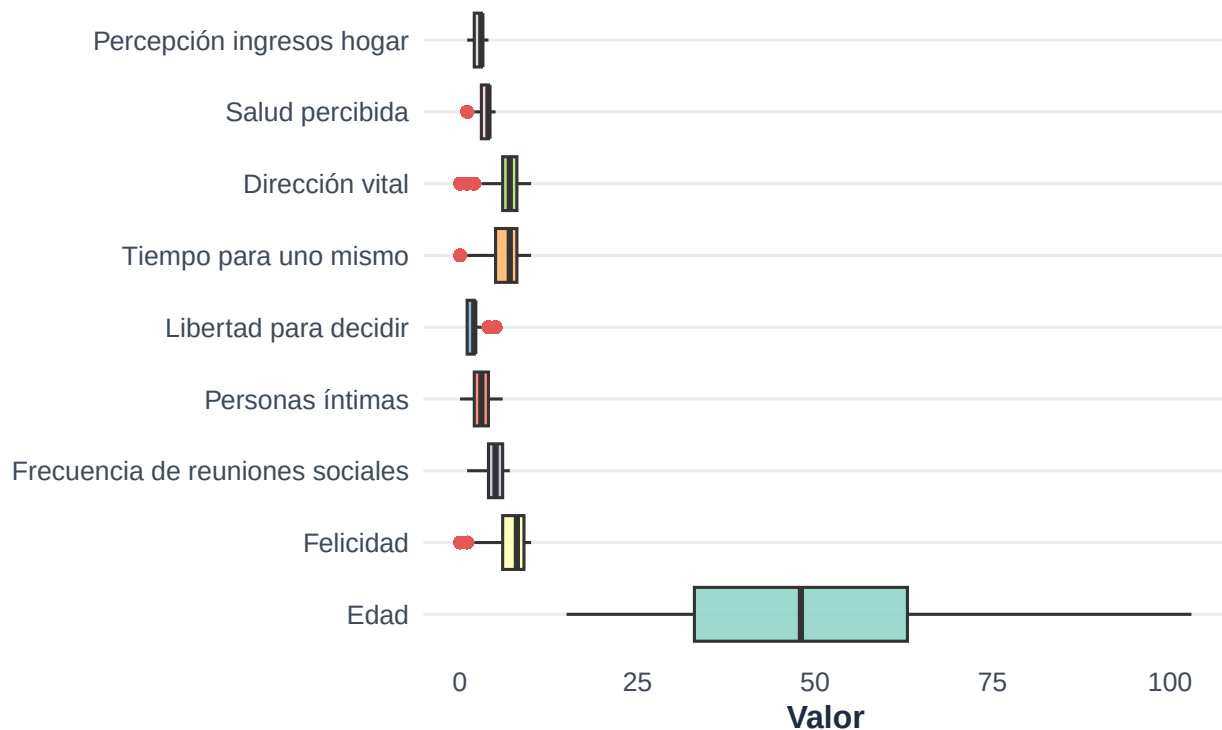
Exploración de forma, concentración y asimetría



g_outliers

Valores atípicos y dispersión

Boxplots de variables clave



```
ggsave("Graficos/01_distribuciones_clave.png", g_dist, width = 12, height = 8, dpi = 300)
ggsave("Graficos/02_outliers_clave.png", g_outliers, width = 10, height = 7, dpi = 300)
```

```
# -----
# 9.5 Relación entre edad y bienestar
# -----

df_edad_bienestar <- df_viz %>%
  select(
    agea,
    all_of(vars_vd_dashboard_num)
  ) %>%
  pivot_longer(
    cols = -agea,
    names_to = "variable",
    values_to = "valor"
  ) %>%
  filter(
    !is.na(agea),
    !is.na(valor)
  ) %>%
  mutate(
    variable = recode(
      variable,
      !!!etiquetas_num
    )
  )
```

```

set.seed(123)

df_edad_bienestar_muestra <- df_edad_bienestar %>%
  group_by(variable) %>%
  group_modify(~{
    n_muestra <- min(3000, nrow(.x))
    slice_sample(.x, n = n_muestra)
  }) %>%
  ungroup()

g_edad_bienestar <- ggplot(
  df_edad_bienestar_muestra,
  aes(
    x = agea,
    y = valor
  )
) +
  geom_point(
    alpha = 0.10,
    color = "#F58518",
    size = 0.8
  ) +
  geom_smooth(
    method = "loess",
    se = FALSE,
    color = "#1F2A44",
    linewidth = 0.8
  ) +
  facet_wrap(
    ~ variable,
    scales = "free_y",
    ncol = 3
  ) +
  labs(
    title = "Relación entre edad e indicadores de bienestar",
    subtitle = "Dispersión con muestra aleatoria y tendencia suavizada",
    x = "Edad",
    y = "Valor"
  )

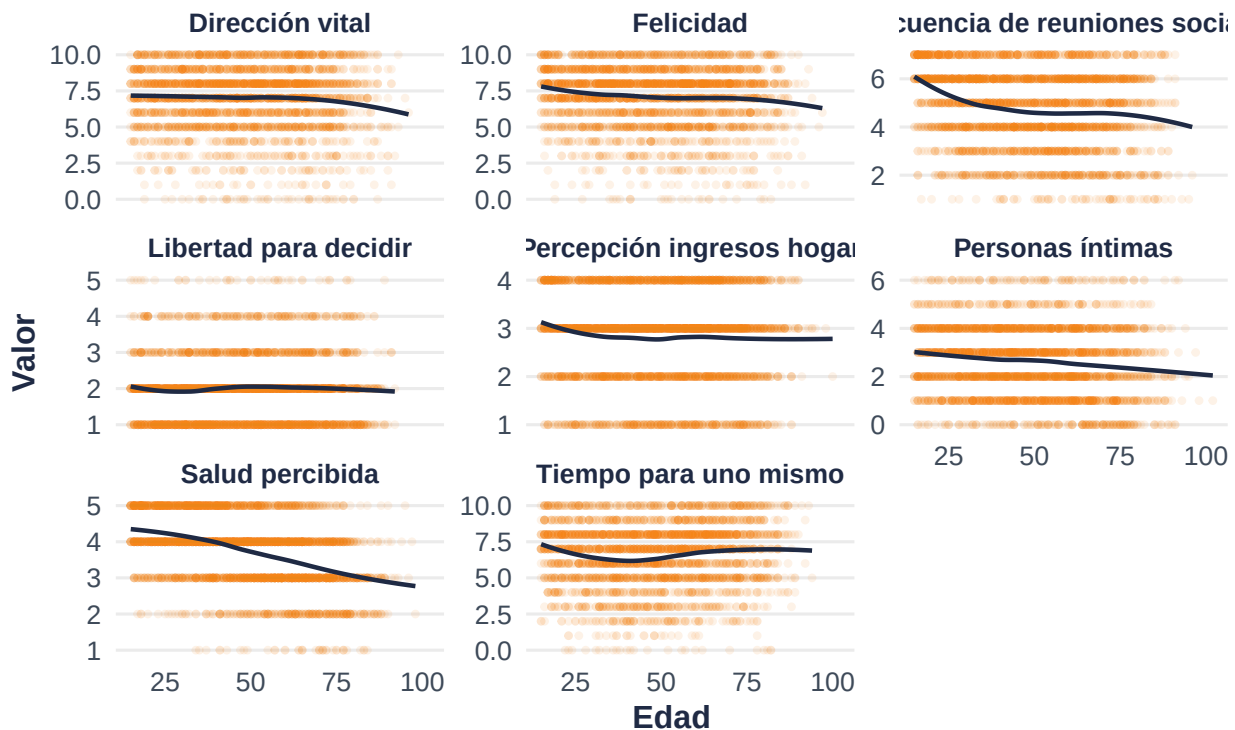
g_edad_bienestar

## `geom_smooth()` using formula = 'y ~ x'

```

Relación entre edad e indicadores de bienestar

Dispersión con muestra aleatoria y tendencia suavizada



```
ggsave(
  "Graficos/03_edad_bienestar.png",
  g_edad_bienestar,
  width = 12,
  height = 8,
  dpi = 300
)

## `geom_smooth()` using formula = 'y ~ x'
# 9.6 Indicadores de bienestar según nivel educativo

vars_bienestar <- c(
  "happy", "sclmeet", "inprdsc", "dclvlf",
  "tmdotwa", "sedirlf", "health_num", "hincfel_num"
)

df_heat_educ <- df_viz %>%

  # ELIMINAR REGISTROS SIN NIVEL EDUCATIVO
  filter(!is.na(eiscd)) %>%

  group_by(eiscd) %>%

  summarise(
    across(
      all_of(vars_bienestar),
      ~ mean(.x, na.rm = TRUE)
    )
  )
```

```

    ),
    .groups = "drop"
  ) %>%

  pivot_longer(
    cols = -eisced,
    names_to = "indicador",
    values_to = "media"
  ) %>%

  mutate(
    indicador = recode(indicador, !!!etiquetas_num)
  )

ggplot(df_heat_educ,
  aes(x = indicador,
    y = eisced,
    fill = media)) +

  geom_tile(color = "white") +

  geom_text(
    aes(label = sprintf("%.1f", media)), # etiquetas numéricas, sin "NA"
    size = 3
  ) +

  scale_fill_gradient(
    low = "#DCEAF6",
    high = "#2C6DA4",
    na.value = "white"
  ) +

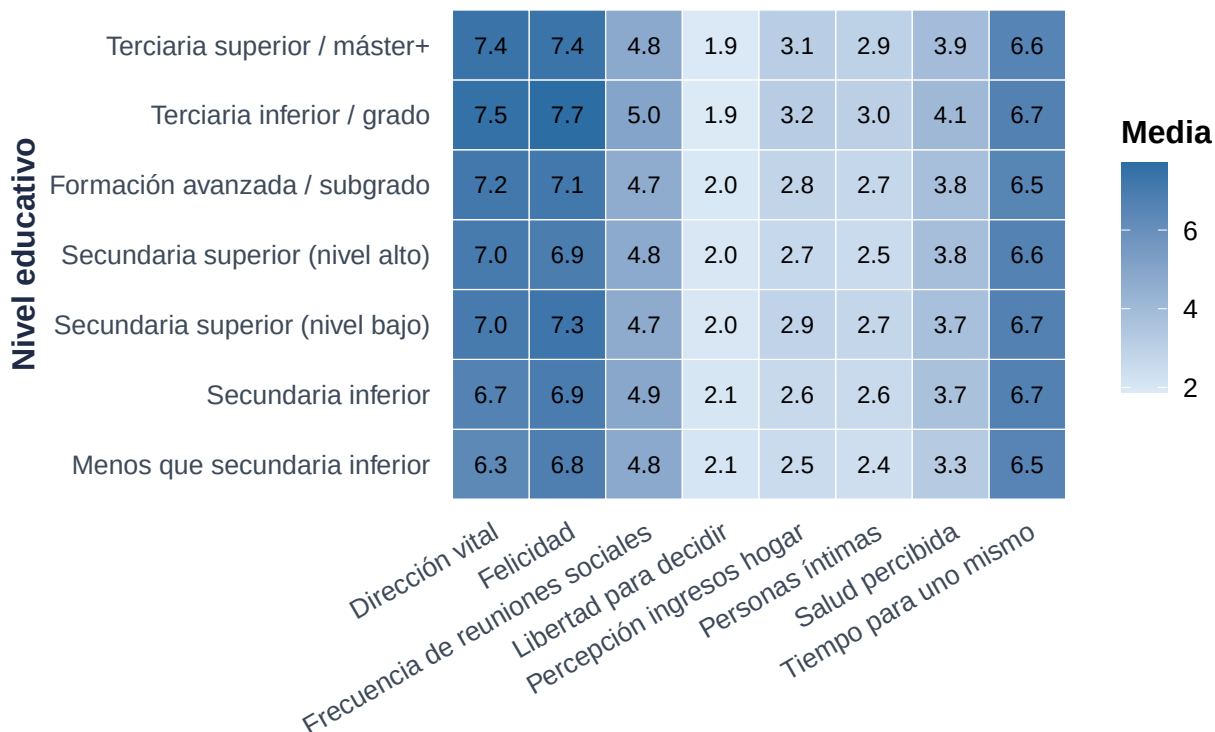
  labs(
    title = "Bienestar medio según nivel educativo",
    subtitle = "Valores medios por nivel educativo",
    x = NULL,
    y = "Nivel educativo",
    fill = "Media"
  ) +

  theme(
    axis.text.x = element_text(
      angle = 30,
      hjust = 1
    ),
    panel.grid = element_blank()
  )

```

Bienestar medio según nivel educativo

Valores medios por nivel educativo



```
# -----
# 9.7 Funciones auxiliares para subgrupos
# -----

pal_salud <- c(
  "Muy mala" = "#08306B",
  "Mala"     = "#2171B5",
  "Regular"  = "#6BAED6",
  "Buena"    = "#C6DBEF",
  "Muy buena" = "#DEEBF7"
)

pal_ingresos <- c(
  "Mucha dificultad" = "#7F0000",
  "Dificultad"       = "#D7301F",
  "Satisface"        = "#FC8D59",
  "Muy satisfecho"   = "#FEE0D2"
)

# -----
# Distribuciones tipo Likert
# -----

graficar_likert_subgrupo <- function(
  data,
  grupo_var,
  respuesta_var,
```

```

respuesta_num_var,
titulo,
xlab,
fill_lab,
paleta
) {

df_plot <- data %>%
  filter(
    !is.na(.data[[grupo_var]]),
    !is.na(.data[[respuesta_var]]),
    !is.na(.data[[respuesta_num_var]])
  ) %>%
  count(.data[[grupo_var]], .data[[respuesta_var]]) %>%
  group_by(.data[[grupo_var]]) %>%
  mutate(prop = n / sum(n)) %>%
  ungroup() %>%
  left_join(
    data %>%
      filter(
        !is.na(.data[[grupo_var]]),
        !is.na(.data[[respuesta_num_var]])
      ) %>%
      group_by(.data[[grupo_var]]) %>%
      summarise(
        media_orden = mean(.data[[respuesta_num_var]], na.rm = TRUE),
        .groups = "drop"
      ),
    by = grupo_var
  ) %>%
  mutate(
    grupo_ordenado = fct_reorder(
      as.factor(.data[[grupo_var]]),
      media_orden
    )
  )

ggplot(
  df_plot,
  aes(
    x = grupo_ordenado,
    y = prop,
    fill = .data[[respuesta_var]]
  )
) +
  geom_col(
    width = 0.72,
    colour = "white"
  ) +
  scale_y_continuous(labels = percent_format()) +
  scale_fill_manual(values = paleta) +
  coord_flip() +
  labs(

```

```

    title = titulo,
    subtitle = "Distribución porcentual por subgrupo",
    x = xlab,
    y = "Porcentaje",
    fill = fill_lab
  )
}

# -----
# Heatmaps de perfiles medios
# -----

graficar_heatmap_perfiles <- function(
  data,
  grupo_var,
  titulo,
  ylab,
  variables_resumen
) {

  resumen <- data %>%
    filter(!is.na(.data[[grupo_var]])) %>%
    group_by(.data[[grupo_var]]) %>%
    summarise(
      across(
        all_of(variables_resumen),
        ~ mean(.x, na.rm = TRUE)
      ),
      .groups = "drop"
    ) %>%
    pivot_longer(
      cols = -all_of(grupo_var),
      names_to = "indicador",
      values_to = "media"
    ) %>%
    filter(!is.na(media)) %>%
    mutate(
      indicador = recode(indicador, !!!etiquetas_num)
    )

  ggplot(
    resumen,
    aes(
      x = indicador,
      y = .data[[grupo_var]],
      fill = media
    )
  ) +
    geom_tile(
      colour = "white",
      linewidth = 0.6
    ) +
    geom_text(

```



```

    aes(label = sprintf("%.1f", media)),
    size = 3.2,
    colour = "#1F2A44"
  ) +
  scale_fill_gradient(
    low = "#DCEAF6",
    high = "#2C6DA4"
  ) +
  labs(
    title = titulo,
    subtitle = "Perfil medio de variables de bienestar",
    x = NULL,
    y = ylab,
    fill = "Media"
  ) +
  theme(
    axis.text.x = element_text(
      angle = 30,
      hjust = 1
    ),
    panel.grid = element_blank()
  )
}

```

```

# -----
# 9.8 Exploración VI -> VD
# -----

# =====
# SEXO
# =====

g_salud_sexo <- graficar_likert_subgrupo(
  data = df_viz,
  grupo_var = "gndr",
  respuesta_var = "health",
  respuesta_num_var = "health_num",
  titulo = "Salud percibida según sexo",
  xlab = "Sexo",
  fill_lab = "Salud percibida",
  paleta = pal_salud
)

g_ingresos_sexo <- graficar_likert_subgrupo(
  data = df_viz,
  grupo_var = "gndr",
  respuesta_var = "hincfel",
  respuesta_num_var = "hincfel_num",
  titulo = "Percepción de ingresos según sexo",
  xlab = "Sexo",
  fill_lab = "Percepción ingresos",
  paleta = pal_ingresos
)

```

```

g_heat_sexo <- graficar_heatmap_perfiles(
  data = df_viz,
  grupo_var = "gndr",
  titulo = "Perfil de bienestar según sexo",
  ylab = "Sexo",
  variables_resumen = vars_vd_dashboard_num
)

# =====
# PAREJA
# =====

g_salud_pareja <- graficar_likert_subgrupo(
  df_viz,
  "icpart1",
  "health",
  "health_num",
  "Salud percibida según convivencia en pareja",
  "Convivencia en pareja",
  "Salud percibida",
  pal_salud
)

g_ingresos_pareja <- graficar_likert_subgrupo(
  df_viz,
  "icpart1",
  "hincfel",
  "hincfel_num",
  "Percepción de ingresos según convivencia en pareja",
  "Convivencia en pareja",
  "Percepción ingresos",
  pal_ingresos
)

g_heat_pareja <- graficar_heatmap_perfiles(
  df_viz,
  "icpart1",
  "Perfil de bienestar según convivencia en pareja",
  "Convivencia en pareja",
  vars_vd_dashboard_num
)

# =====
# HIJOS
# =====

g_salud_hijos <- graficar_likert_subgrupo(
  df_viz,
  "chldhhe",
  "health",
  "health_num",
  "Salud percibida según presencia de niños en el hogar",
  "Niños en el hogar",

```

```

    "Salud percibida",
    pal_salud
  )

g_ingresos_hijos <- graficar_likert_subgrupo(
  df_viz,
  "chldhhe",
  "hincfel",
  "hincfel_num",
  "Percepción de ingresos según presencia de niños en el hogar",
  "Niños en el hogar",
  "Percepción ingresos",
  pal_ingresos
)

g_heat_hijos <- graficar_heatmap_perfiles(
  df_viz,
  "chldhhe",
  "Perfil de bienestar según presencia de niños",
  "Niños en el hogar",
  vars_vd_dashboard_num
)

# =====
# DIVORCIO
# =====

g_salud_divorcio <- graficar_likert_subgrupo(
  df_viz,
  "dvrdeva",
  "health",
  "health_num",
  "Salud percibida según antecedente de divorcio",
  "Antecedente de divorcio",
  "Salud percibida",
  pal_salud
)

g_ingresos_divorcio <- graficar_likert_subgrupo(
  df_viz,
  "dvrdeva",
  "hincfel",
  "hincfel_num",
  "Percepción de ingresos según antecedente de divorcio",
  "Antecedente de divorcio",
  "Percepción ingresos",
  pal_ingresos
)

g_heat_divorcio <- graficar_heatmap_perfiles(
  df_viz,
  "dvrdeva",
  "Perfil de bienestar según antecedente de divorcio",

```

```

"Antecedente de divorcio",
vars_vd_dashboard_num
)

# =====
# EDUCACIÓN
# =====

g_salud_educ <- graficar_likert_subgrupo(
  df_viz,
  "eisced",
  "health",
  "health_num",
  "Salud percibida según nivel educativo",
  "Nivel educativo",
  "Salud percibida",
  pal_salud
)

g_ingresos_educ <- graficar_likert_subgrupo(
  df_viz,
  "eisced",
  "hincfel",
  "hincfel_num",
  "Percepción de ingresos según nivel educativo",
  "Nivel educativo",
  "Percepción ingresos",
  pal_ingresos
)

g_heat_educ <- graficar_heatmap_perfiles(
  df_viz,
  "eisced",
  "Perfil de bienestar según nivel educativo",
  "Nivel educativo",
  vars_vd_dashboard_num
)

# =====
# INGRESOS COMO VARIABLE INDEPENDIENTE
# =====

g_salud_ingresos <- graficar_likert_subgrupo(
  df_viz,
  "hincfel",
  "health",
  "health_num",
  "Salud percibida según ingresos del hogar",
  "Percepción ingresos hogar",
  "Salud percibida",
  pal_salud
)

```

```

g_heat_ingresos <- graficar_heatmap_perfiles(
  df_viz,
  "hincfel",
  "Perfil de bienestar según ingresos del hogar",
  "Percepción ingresos hogar",
  setdiff(vars_vd_dashboard_num, "hincfel_num")
)

# -----
# 9.9 Mostrar y guardar gráficas
# -----

graficos_lista <- list(

  g_salud_sexo = g_salud_sexo,
  g_ingresos_sexo = g_ingresos_sexo,
  g_heat_sexo = g_heat_sexo,

  g_salud_pareja = g_salud_pareja,
  g_ingresos_pareja = g_ingresos_pareja,
  g_heat_pareja = g_heat_pareja,

  g_salud_hijos = g_salud_hijos,
  g_ingresos_hijos = g_ingresos_hijos,
  g_heat_hijos = g_heat_hijos,

  g_salud_divorcio = g_salud_divorcio,
  g_ingresos_divorcio = g_ingresos_divorcio,
  g_heat_divorcio = g_heat_divorcio,

  g_salud_educ = g_salud_educ,
  g_ingresos_educ = g_ingresos_educ,
  g_heat_educ = g_heat_educ,

  g_salud_ingresos = g_salud_ingresos,
  g_heat_ingresos = g_heat_ingresos
)

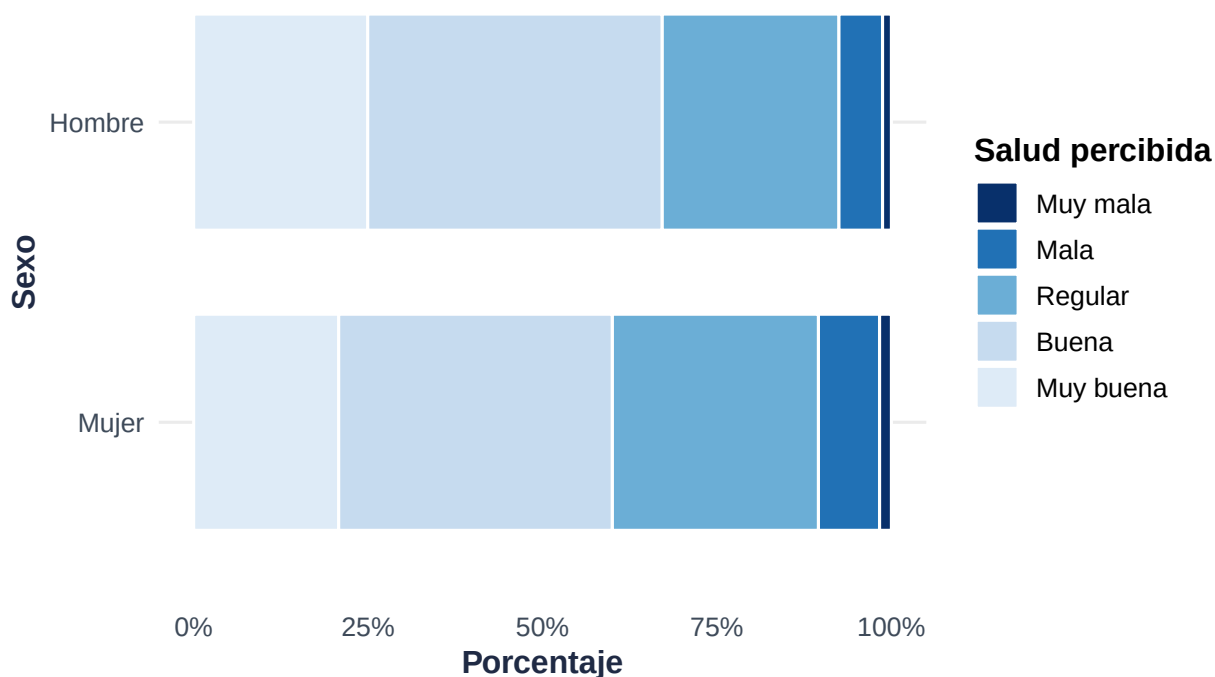
# Mostrar todas las gráficas

walk(
  graficos_lista,
  print
)

```

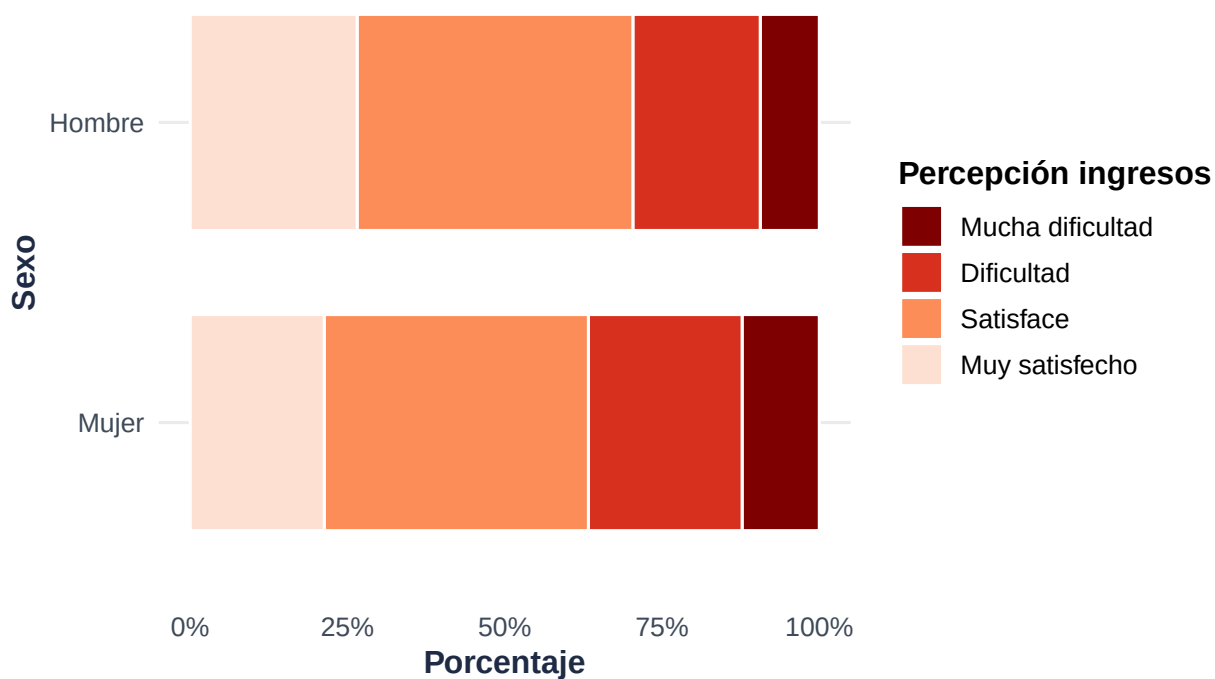
Salud percibida según sexo

Distribución porcentual por subgrupo



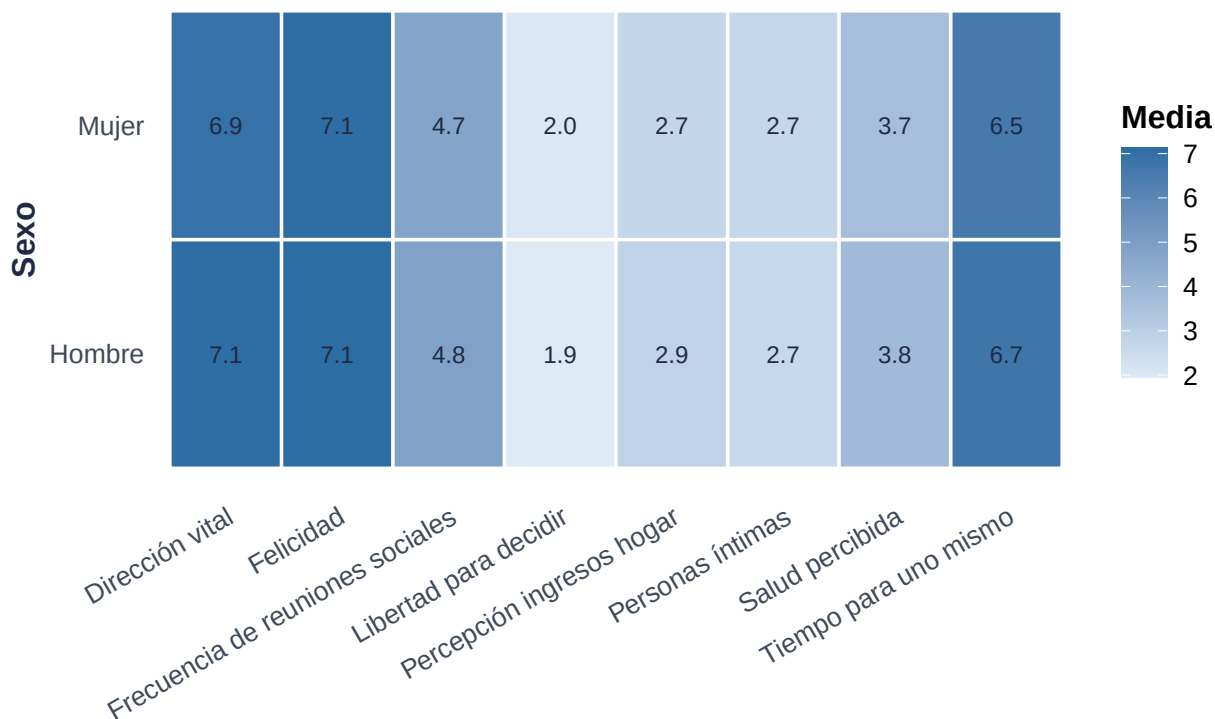
Percepción de ingresos según sexo

Distribución porcentual por subgrupo



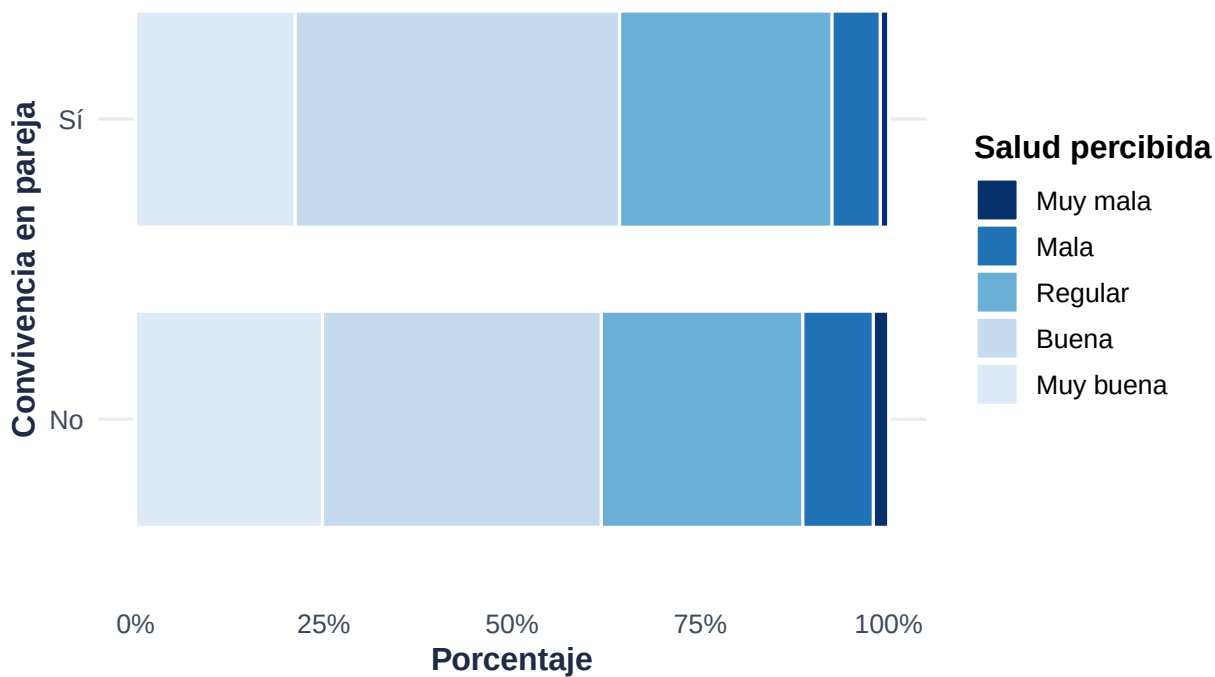
Perfil de bienestar según sexo

Perfil medio de variables de bienestar



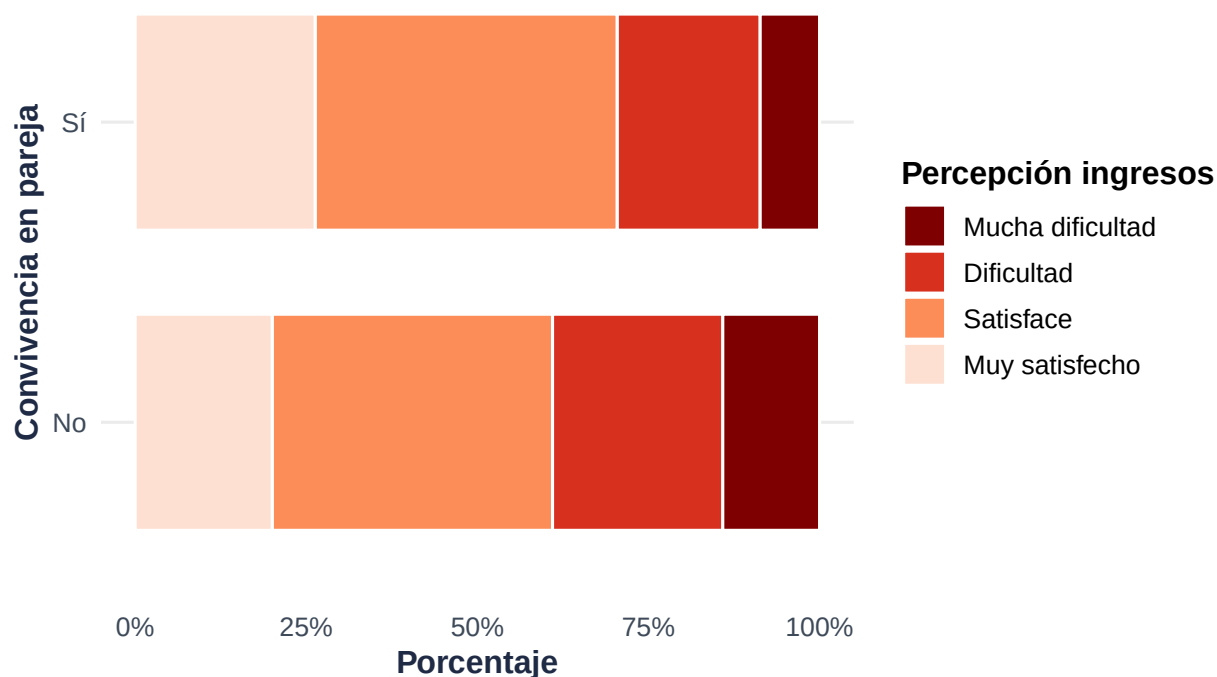
Salud percibida según convivencia en pareja

Distribución porcentual por subgrupo



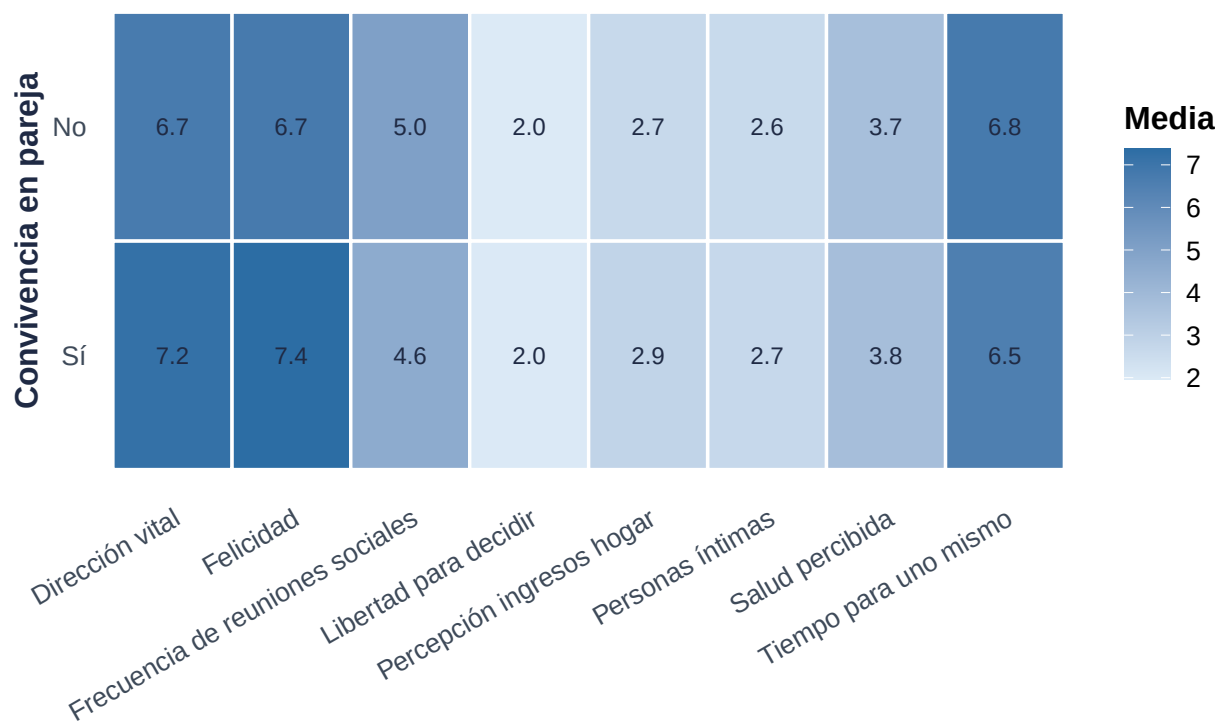
Percepción de ingresos según convivencia en pareja

Distribución porcentual por subgrupo



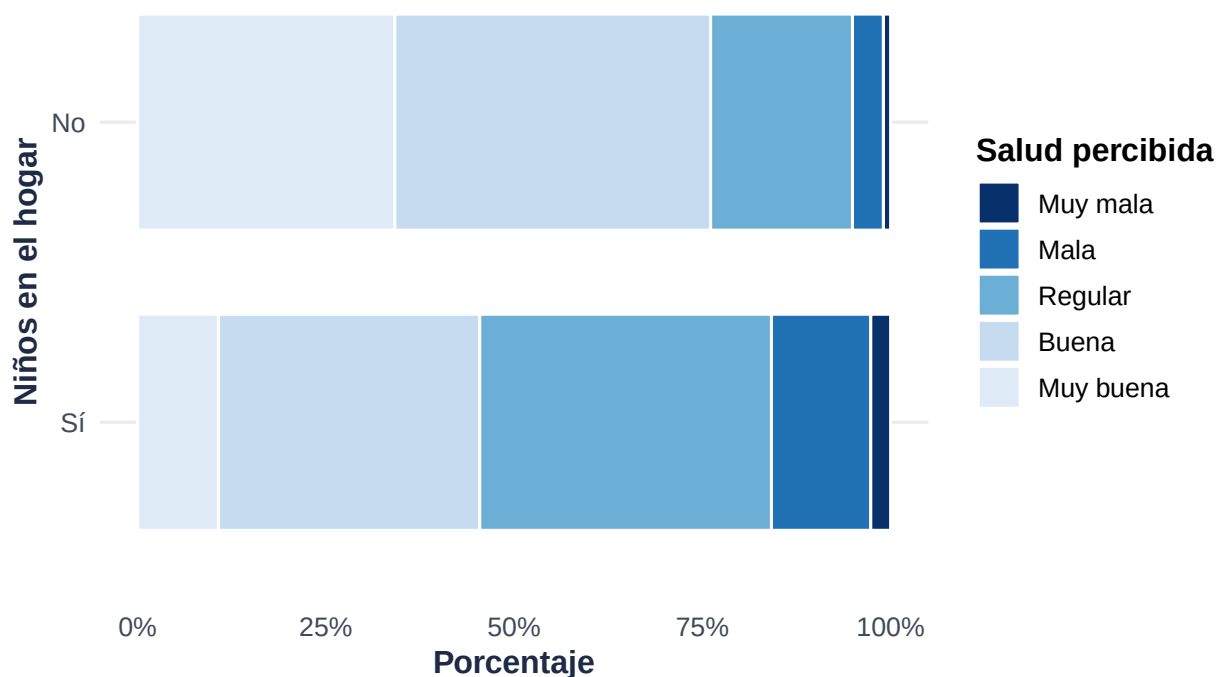
Perfil de bienestar según convivencia en pareja

Perfil medio de variables de bienestar



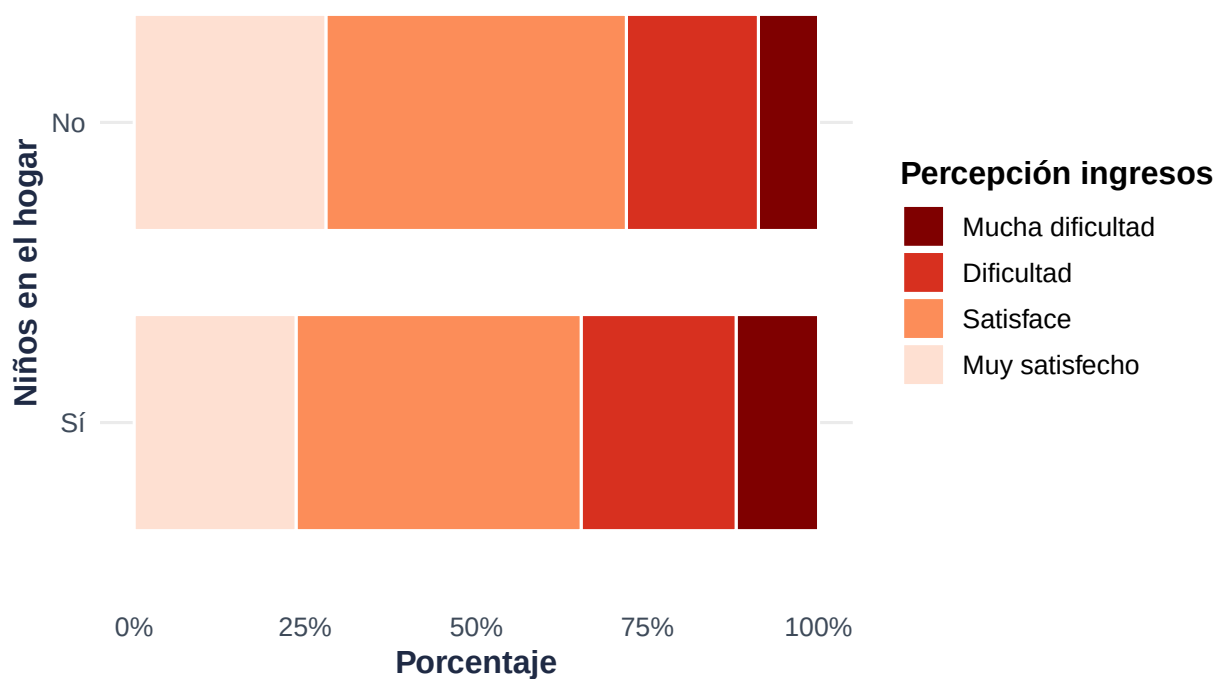
Salud percibida según presencia de niños en el hogar

Distribución porcentual por subgrupo



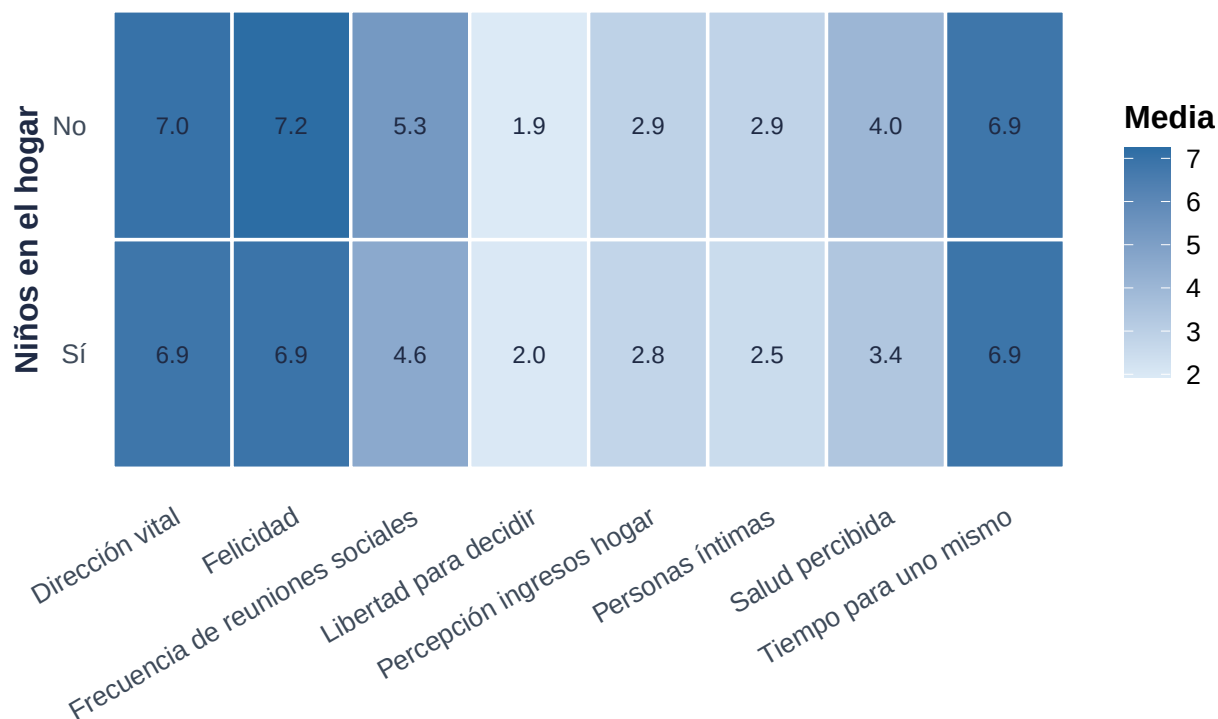
Percepción de ingresos según presencia de niños en el hogar

Distribución porcentual por subgrupo



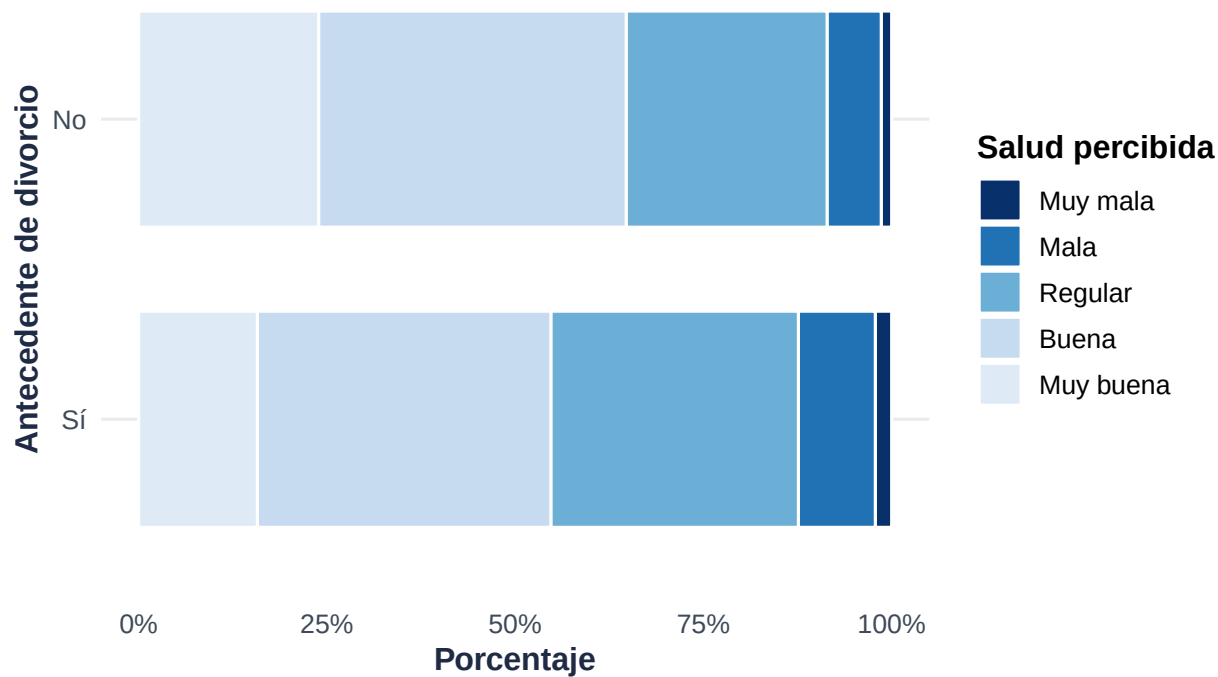
Perfil de bienestar según presencia de niños

Perfil medio de variables de bienestar



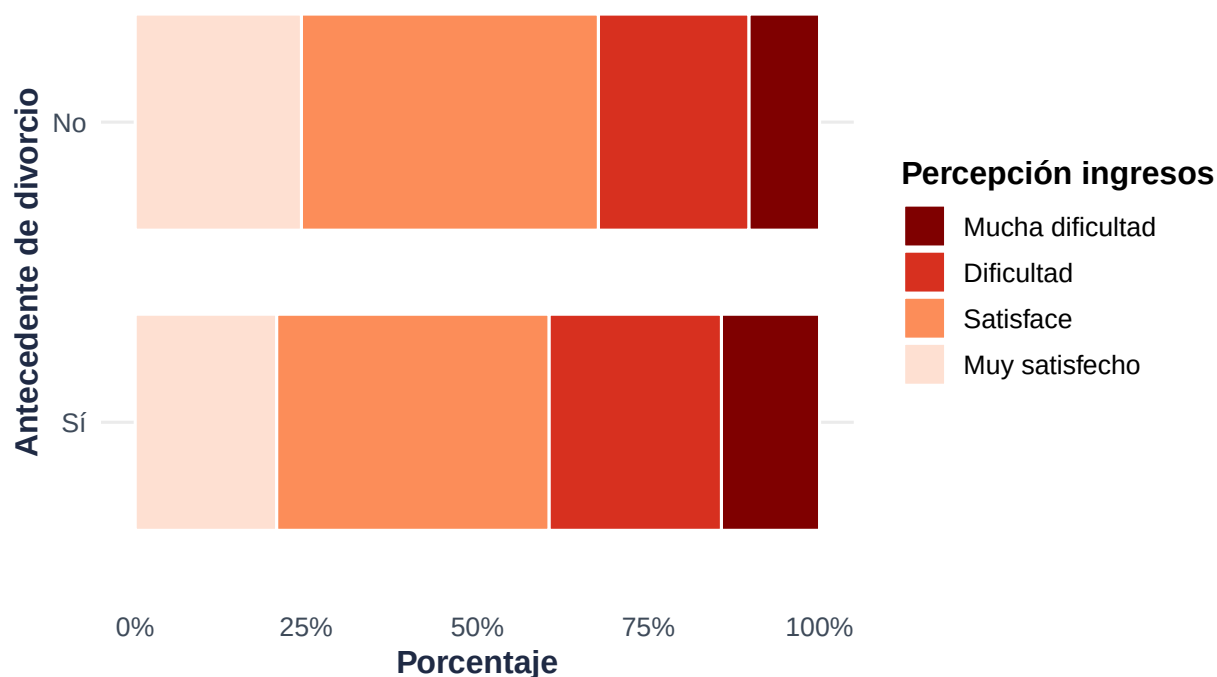
Salud percibida según antecedente de divorcio

Distribución porcentual por subgrupo



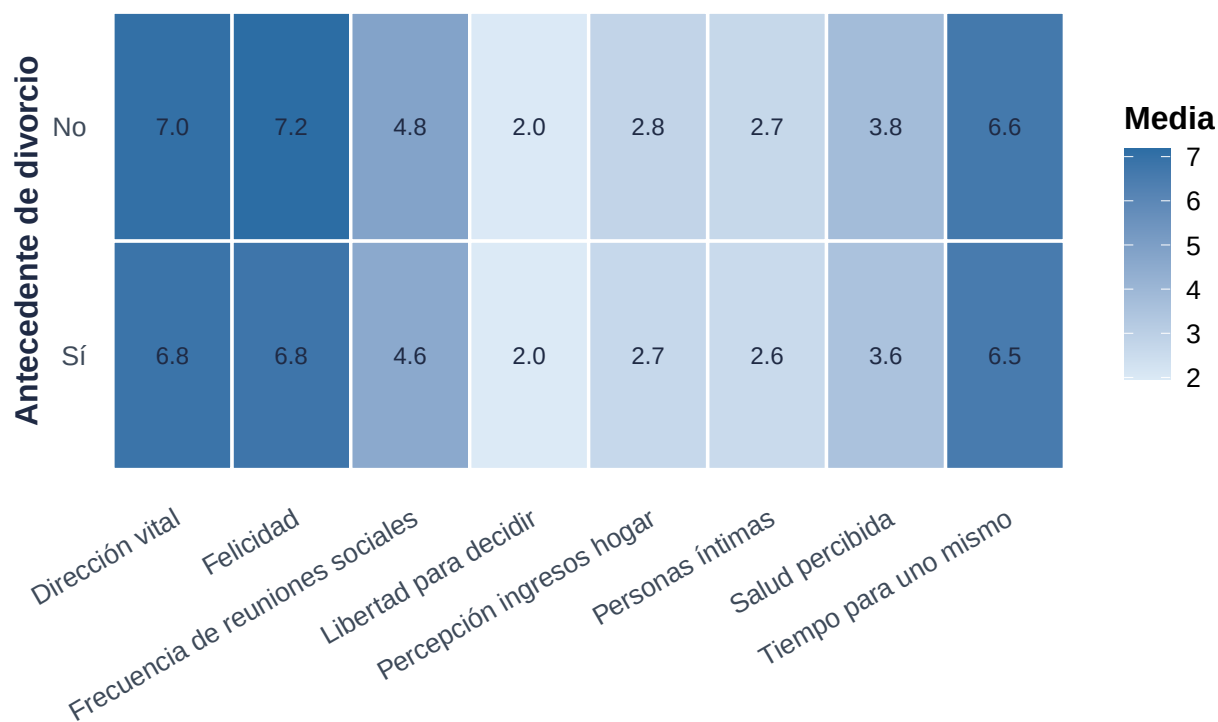
Percepción de ingresos según antecedente de divorcio

Distribución porcentual por subgrupo



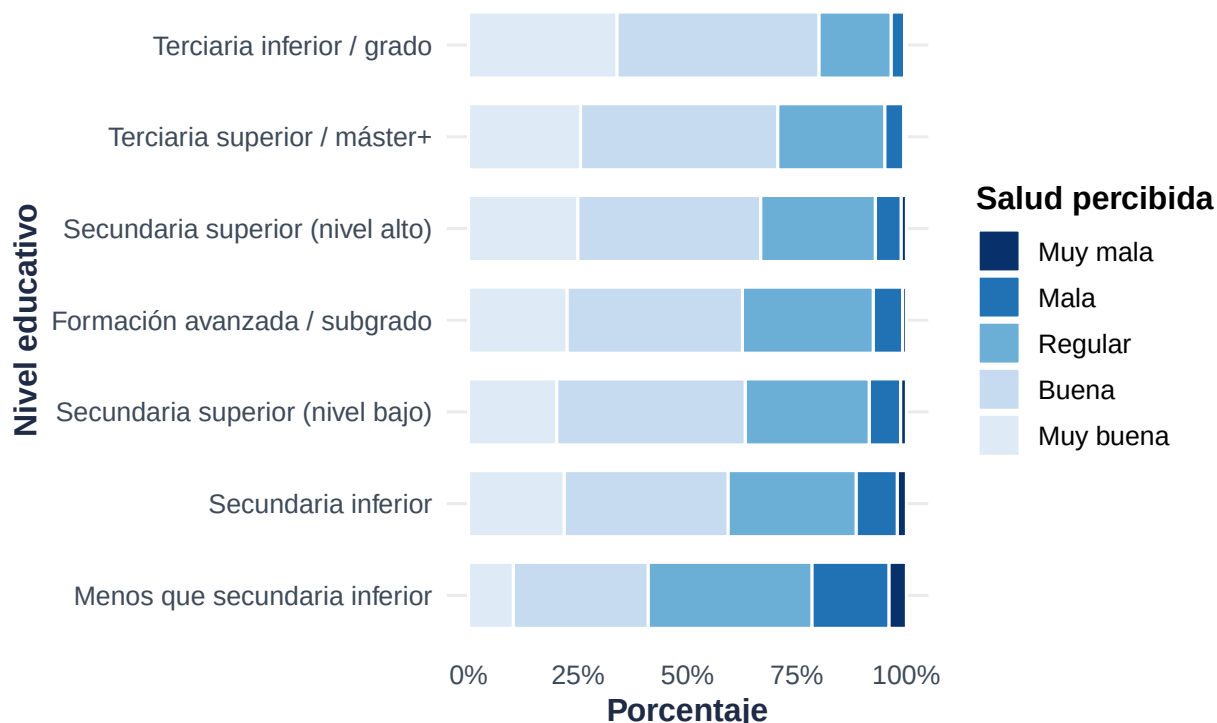
Perfil de bienestar según antecedente de divorcio

Perfil medio de variables de bienestar



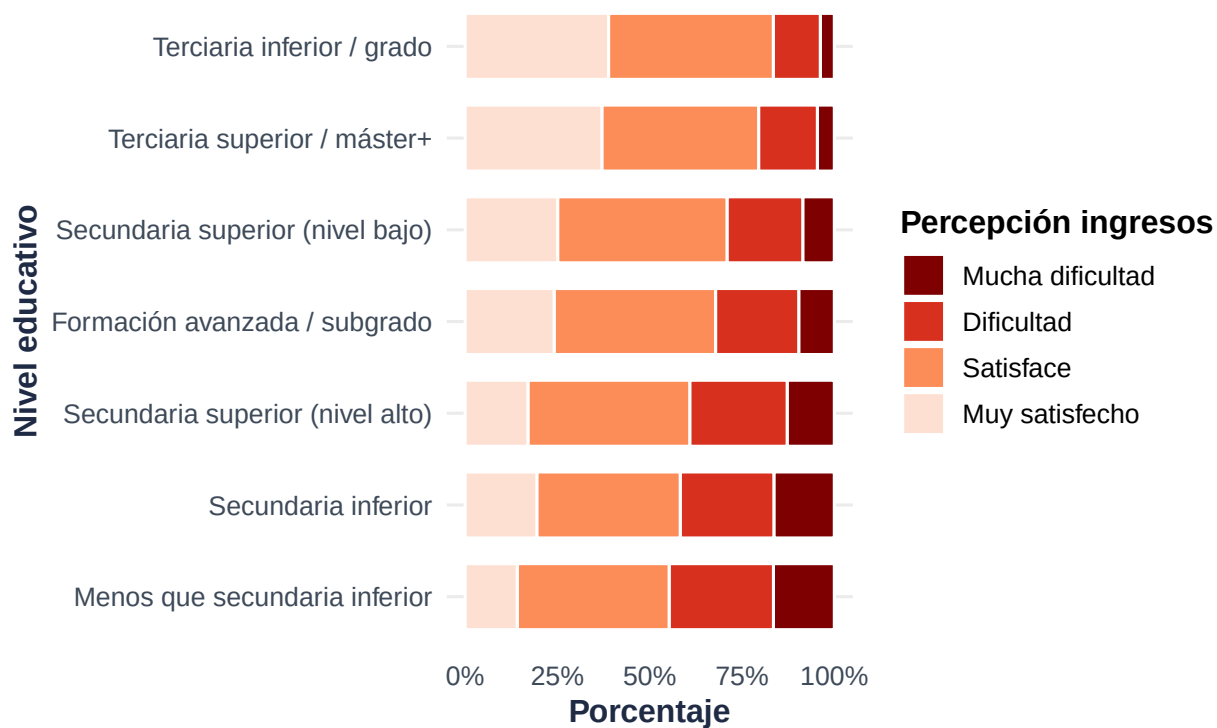
Salud percibida según nivel educativo

Distribución porcentual por subgrupo



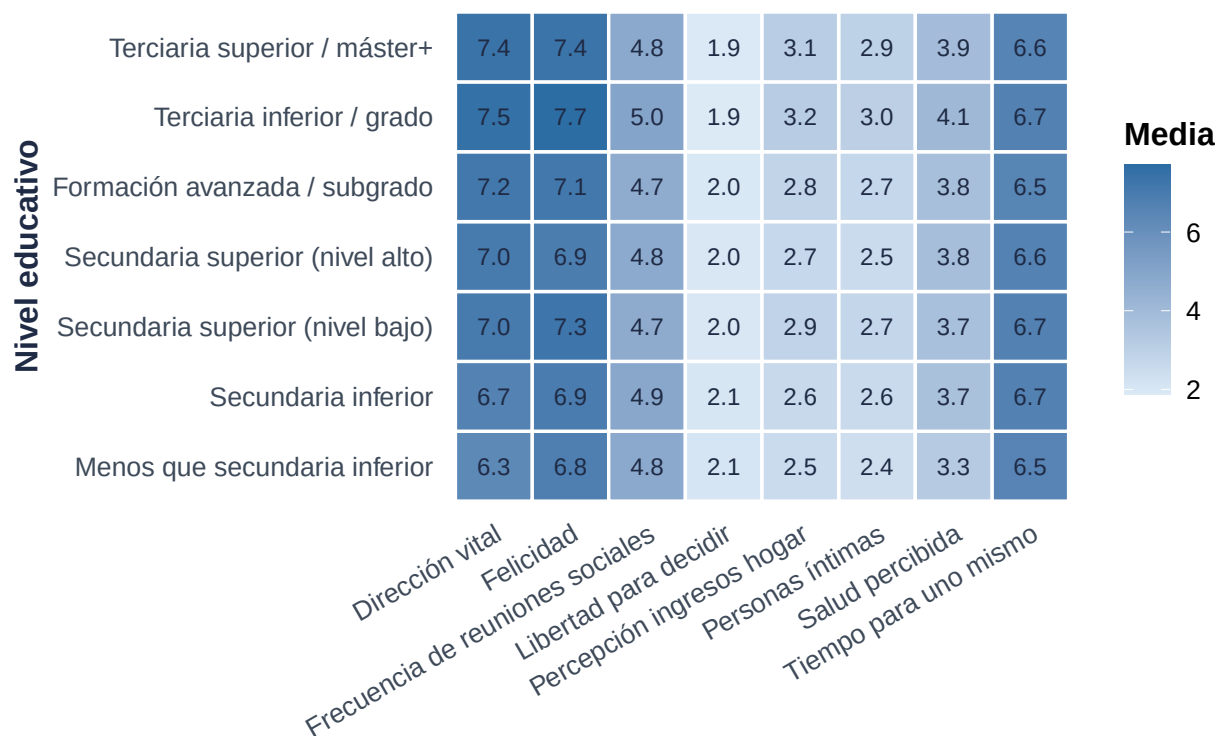
Percepción de ingresos según nivel educativo

Distribución porcentual por subgrupo



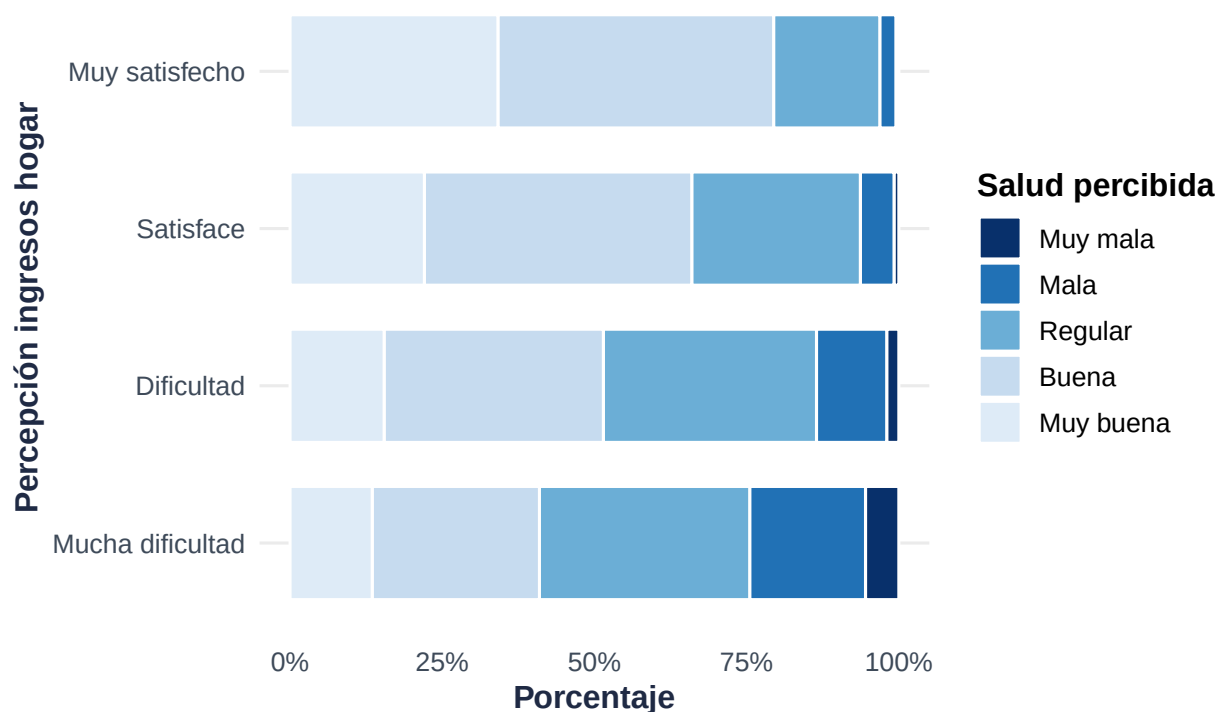
Perfil de bienestar según nivel educativo

Perfil medio de variables de bienestar



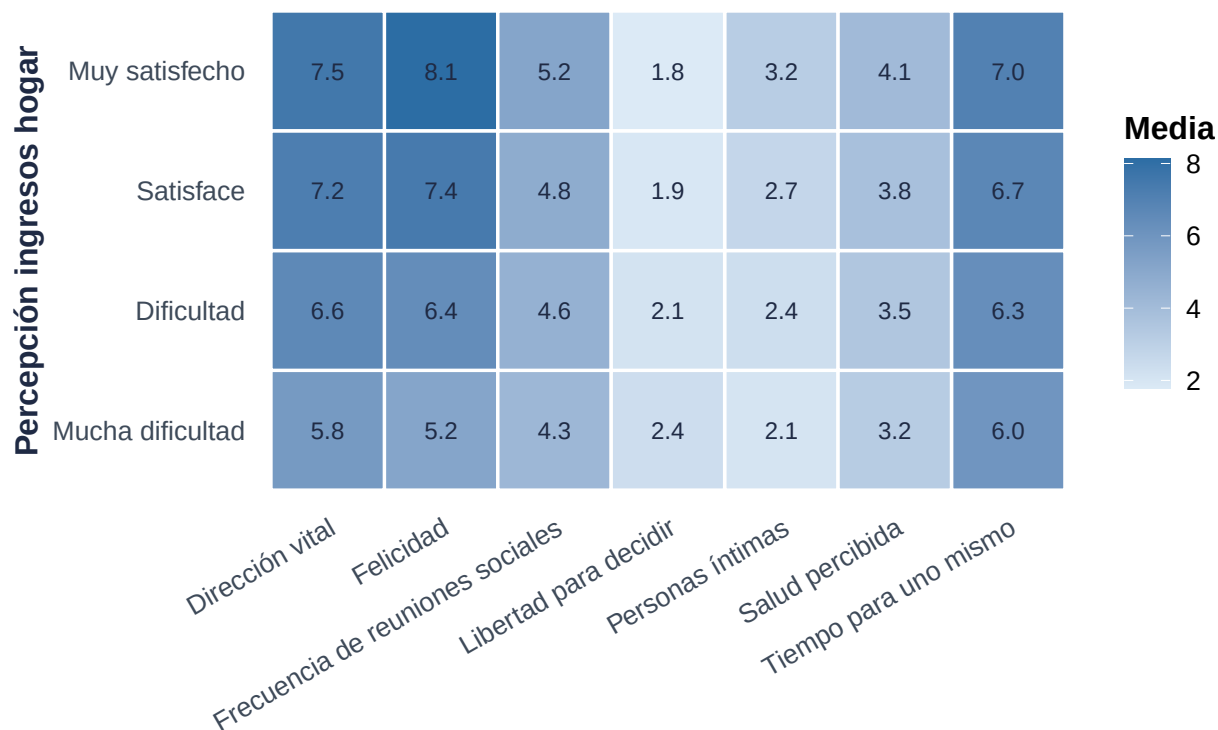
Salud percibida según ingresos del hogar

Distribución porcentual por subgrupo



Perfil de bienestar según ingresos del hogar

Perfil medio de variables de bienestar



Guardar todas las gráficas

```
walk2(
  .x = graficos_lista,
  .y = names(graficos_lista),
  .f = ~ ggsave(
    filename = file.path(
      "Graficos",
      paste0(.y, ".png")
    ),
    plot = .x,
    width = 12,
    height = 8,
    dpi = 300
  )
)
```